

Air outlets

In this section, we will explain the air outlets used in HVAC systems. Air outlets are one of the most important components of an HVAC system in buildings and facilities. Their most important functions are:

- Distributing air in the room in a suitable pattern to ensure proper distribution and increase cooling efficiency
- Direct the air path in the appropriate pattern within the room (To ensure its proper distribution and thus increase cooling efficiency)
- Preventing the transfer of unwanted objects.
- Controlling air flow rate
- Decorative purposes

Important Components of HVAC Systems /Air Outlets of HVAC System:

-Grilles and Diffusers:

Grilles, diffusers and louvers are essential components of HVAC systems, responsible for distributing conditioned air throughout a space. They play a crucial role in maintaining a comfortable and healthy indoor environment.

- Grilles and diffusers play a vital role in HVAC systems.
- Grilles provide direct airflow, while diffusers direct air at different angles for better mixing.

Grilles vs. Diffusers Vs. Louvers

- **Grilles:** Allow air to pass through in a straight line, typically used in applications requiring a long throw of air, such as hallways or large open spaces. (More suitable for installation in walls)
- **Diffusers:** Modify the direction of air as it exits the terminal unit, promoting better air mixing and creating a more uniform temperature distribution. Commonly used in occupied spaces. (More suitable for installation in Ceilings).
- **Louvers:** - Act as safeguard against the intrusion of undesirable substances such as Rain & Dust, they are usually, accompanied by filters. Their prevalent use in exterior vents, air intakes, and exhaust vents underscores their importance in maintaining air quality and preventing the entry of contaminants. (For Outdoor/ exterior Installation).

Understanding the Different Types of Air Outlets

When choosing air outlets, it's crucial to understand the distinctions between the primary types: supply, return, dummy, Fresh, Exhaust and Transfer.

Supply Air Outlets:

Supply air outlets are responsible for distributing conditioned air from the HVAC system into the room. They are equipped with both dampers and deflectors to regulate airflow and direct it effectively. Dampers control the volume of air passing through the outlet, while deflectors direct the airflow to achieve the desired distribution pattern.

Return Air Outlets:

Return air outlets, unlike supply outlets, do not require dampers or deflectors. Their primary function is to extract stale air from the room and return it to the HVAC system for recirculation. Since there's no need for airflow control or direction, these outlets typically have a simpler design.

Dummy Air Outlets:

Dummy air outlets, as the name suggests, serve no functional purpose. They are primarily used to maintain aesthetic consistency and comply with design requirements. They often resemble as return outlets but are internally blocked (with blank sheet) to prevent airflow.

Exhaust Air Outlets:

These outlets are vital for removing unwanted air from the building, eliminating pollutants, fumes, and unpleasant odors. Their design and construction vary depending on their application:

- **Indoor Application:** Typically share a similar design as return air outlets (no dampers or deflectors), exhaust outlets can incorporate these features (similar to supply air outlets) for specific applications like fume hoods or labs where controlled exhaust is crucial.
- **Outdoor Application:** Louvers are designed to direct the exhaust air away from the building and surrounding areas. Some outdoor exhaust air outlets (Exhaust Louvers) may incorporate additional features like bird screens, filters or fire dampers.

Fresh Air Outlets:

These outlets introduce fresh, clean air from outdoors into the building, providing improved ventilation and air quality. Their design often incorporates elements similar to supply air outlets:

- **Indoor Application:** Typically share similarities with supply air outlets, utilizing dampers and deflectors for precise airflow control. However, they may also have alternative configurations, like ceiling-mounted diffusers or wall-mounted grilles, to adapt to various needs within the building.
- **Outdoor Application:** Commonly use sand trap louvers to prevent sand and debris from entering the building along with the fresh air. While some may use standard fresh air louvers, many incorporate built-in insect screens to keep pests at bay. Additionally, these outdoor outlets frequently incorporate filters to remove dust and debris from the incoming air, and dampers to ensure precise control over the airflow. This combination of features (Louver + Filter + Damper) ensures clean and healthy fresh air is delivered into the building, creating a comfortable and enjoyable indoor environment.

Transfer Air Outlets:

These outlets serve a specific purpose of transferring air between different spaces within the building without connecting to the main ducting system. They offer several benefits:

Applications:

- Maintaining air pressure balance between different areas.
- Providing ventilation in enclosed spaces without ductwork.
- Transferring air between rooms with different temperatures.

Examples:

- Door grilles
- Access door grilles
- Air grilles with fixed blades

Each type of air outlet plays a crucial role in maintaining optimal air quality, comfort, and ventilation within a building. Choosing the appropriate type and design for each application ensures efficient air distribution and environmental control, ultimately contributing to a healthy and comfortable indoor environment.

Certifications:

In the world of HVAC, certifications play a vital role in ensuring the quality and safety of products. These independent assessments provide assurance to consumers that the products they purchase meet rigorous standards and perform as intended. Here's a breakdown of some key certifications and their significance:

INTERTEK



AHRI



UL



AMCA



ISO



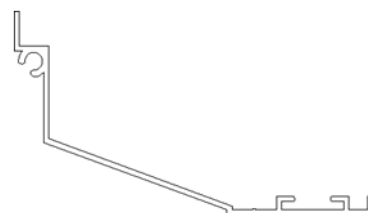
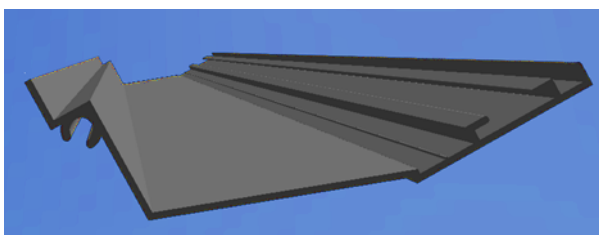
Our products undergo rigorous testing and certification by independent third-party organizations, such as INTERTEK, AMCA, and UL. These certifications provide assurance to our customers that our products meet the highest standards of safety, quality, and performance.

Colors & Finishes:

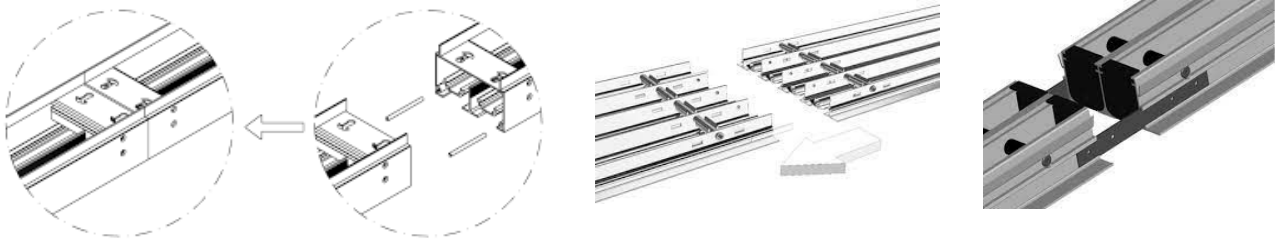
Our air outlets typically come in a standard Traffic White powder coating (RAL 9016), but other colors can be made available upon request. For reference, please scan the provided QR code to view the RAL color chart.



Extrusion Aluminum: In the extrusion process, hot aluminum is pressed through a groove carved in a mold so that the metal section takes the shape of the groove in the end. Therefore, the shape of the section cannot be changed afterwards except by cutting.



The standard length of an aluminum section is 6 meters, so an air outlet frame or blade cannot be manufactured longer than that as a single piece.



IF the length required is more than 6 meters, the diffusers/ grill come in 2 or more pieces and connected together by alignment strips (butt joints) as clarified. You can Scan the QR Code For more clarifications.

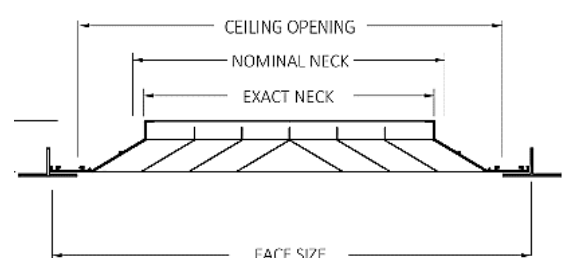


Air outlets consist of the following components:

- **Frame:** Defines neck size and face size.
- **Core:** A set of blades connected to the frame.
- **Damper:** Used to adjust the amount of air before it comes out of the air outlet. It is often used only for supply air, not for return air. The most common type of damper is the opposed blade damper (OBD), which consists of blades that move rotationally (each blade rotates in the opposite direction of the blade next to it) so that it controls the air inlet area and thus the volume flow rate. The damper is usually black so that it is hidden from the outside.
- **Accessories:** Other accessories may be installed, which will be mentioned later.
- The outlet components are typically made of extruded aluminum.

Air outlet dimensions

- **Face size:** This is the largest size of the frame that appears in the end to the room occupant after the diffuser is installed.
- **Neck size:** This is the size of the hidden frame that is installed on the duct or wall opening, etc. Therefore, it is the standard and basic dimension on which air outlets are chosen.
- **Exact Neck Size Vs Nominal Neck Size:** Nominal size is the size of an object that is specified in its name or designation, while exact size is the actual of the product. A 1cm clearance around the actual neck size is typically considered during the design phase to facilitate easier installation. It is important to be aware of the difference between nominal and exact size if you need to make precise measurements.
- **The face size depends on the neck size,** and the difference between them is a fixed amount because as mentioned it is aluminum extrusion that can only be modified by cutting. Therefore, the flange can be cut from the face to get a new face size to reduce the difference between it and the neck size (In this Case it will be a Customized (Non-Standard) and will require additional price.



How to order air outlets from the customer?

- First, the type of diffuser is selected.
- Second, the neck size (standard dimension) is selected based on the size of the opening to be installed on it.
- The diffuser is manufactured so that the actual neck size is smaller than the standard dimension by a certain clearance of approximately 10 mm for easy installation on site. In some cases, the customer specifies the actual neck size, and in this case, this must be clearly stated in the manufacturing order.
- In some cases, the customer orders the diffuser by the actual face size and not the neck size as in the case of square air windows that are placed directly on the false ceiling.

Performance study

Air outlets have a direct impact on the physical properties of the air as it flows through them. Therefore, it is important to consider studying these properties. Here is a simplified explanation of them:

- **Air flow rate:** Air flow rate is the amount of air that flows through the diffuser in a given time. The most common unit of air flow rate is **CFM**, which stands for Minute per Feet Cubic, or cubic feet per minute. When it is mentioned that the flow rate is, for example, CFM 1000, this means that the volume of air flowing in one minute is 1000 cubic feet. There are other well-known units of use such as **LPS**, which stands for Liter per Second or is written in this format L/S, which is one liter per second. Also **m³/hr**, which is cubic meters per hour, etc...
- **Air speed:** Air speed is the distance that air travels in a given time. The most common units are FPM, which stands for Minute Per Feet, meaning feet per minute. Also, m/s, which is meters per second.
- **Free space for the diffuser:** This is the area of the gaps between the blades, which represents the actual area through which air passes through the diffuser.
- **Pressure drop:** Each cooling or fan machine has a certain amount of output pressure, and this pressure decreases due to several factors to reach the conditioned space with the appropriate value. Therefore, the contractor must ensure that the output pressure of the machine is greater than the pressure drop along the duct with its contents, including air outlets, so the pressure drop of the diffusers must be studied.
- **Noise level (NC) criteria:** When air passes through the diffuser blades, the free space of the diffuser is smaller than the duct section and therefore the speed increases and the noise level increase.
- **Throw:** This is the distance that the air will travel from the diffuser before its velocity drops to a specified level. It is important to select a diffuser that has a throw that is appropriate for the size of the room. (For more explanation you can scan this QR code).
 - There is Usually 2 Values for throw one measured vertically (Suitable for wall installation) and one measured Horizontally (Suitable for Ceiling).
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1.Linear Slot Diffusers

Linear Slot Diffusers

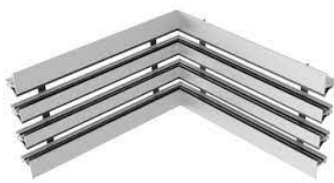
Linear slot diffusers are a type of air diffuser that is becoming increasingly popular in commercial and industrial buildings. They are known for their adaptability and flexibility, as they can be installed in a variety of configurations to meet the specific needs of a space.

Adaptability & Flexibility

Linear slot diffusers come in a variety of lengths and widths, making them suitable for use in a wide range of spaces. They can be installed in straight lines, curved lines, or even in corners. This makes them ideal for use in long corridors, hallways, and other spaces with irregular layouts.

Linear slot diffusers can be installed in both walls and ceilings. This makes them a versatile option for any space. They can be used to provide both supply air and exhaust air. They can be of Flush type or hidden type.

1. **Corner Diffusers:** Corner diffusers are designed to be installed in corners, allowing for efficient air distribution in spaces with limited wall or ceiling area.



Corner wall Inside



Corner wall outside



Corner Ceiling

2. **Straight Diffusers:** Straight diffusers are the most common type and are typically installed along straight walls or ceilings. They can be configured in continuous runs or individual units.



Hidden type-Wall Mounted



Hidden Type Ceiling Mounted



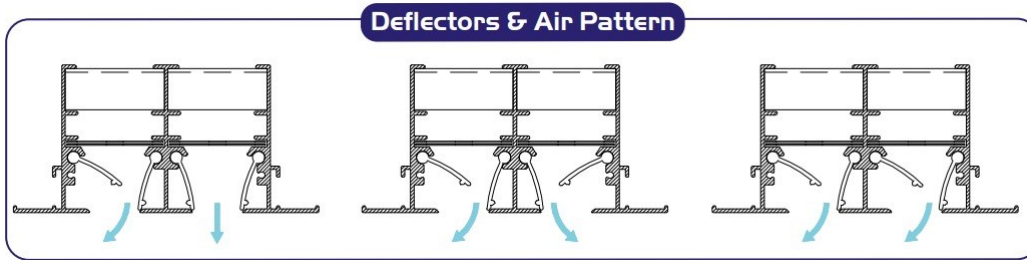
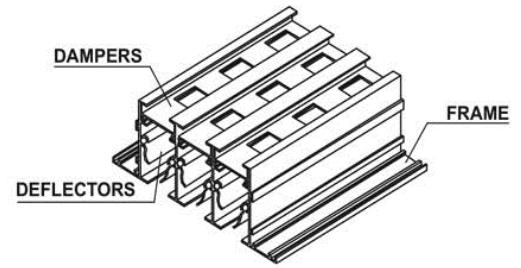
Flush type (Standard).

3. **Curved Diffusers:** Curved diffusers are designed to follow curved architectural elements, such as curved walls or ceilings. They provide seamless airflow distribution in spaces with non-linear layouts.



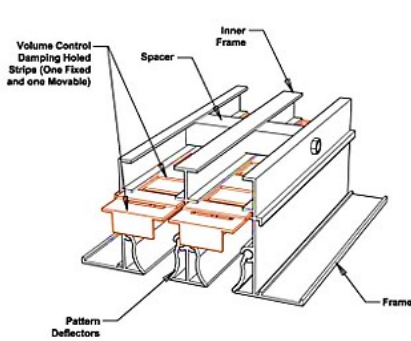
Construction and Components:

- Frame:** Extruded aluminum profiles of 1.1 mm thickness & 17.5 mm flange from each side.
- Deflector & Blades:** Blades form the core of the diffuser, directing the airflow in a specific direction and pattern. They are typically made of extruded aluminum and spacing depending on the intended airflow characteristics.



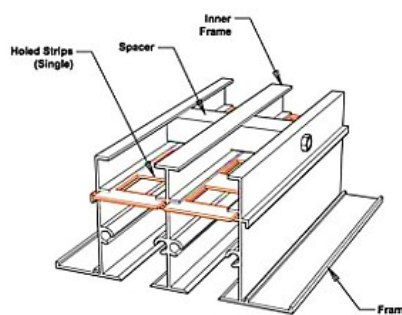
Deflector control mechanisms enable fine-tuning of the airflow distribution. These mechanisms can be manipulated to alter the airflow angle, throw distance, and volume. You can Scan the QR Code for more clarifications.

- Damper Control.** Hit-and-miss dampers are simple yet effective airflow control devices consisting of two holed strips of steel (black color), one fixed and one movable to allow volume control dampening. They are typically controlled manually allowing airflow to pass through when open and blocking it when closed.



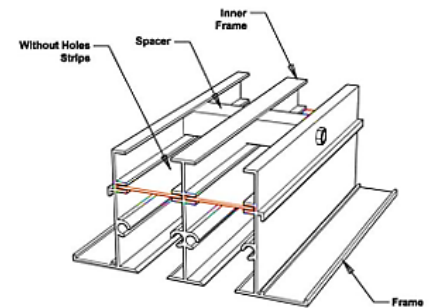
Supply.

Supplied with the 2 strips the instead for air flow control.



Return.

supplied with one fixed holed strip (No control required).



Dummy.

Supplied with Blank Sheet of damper to block air path.

- SLOT WIDTH:** We have a wide range of Slot Sizes (1-8) Slots and 2 standard Slot openings $\frac{3}{4}$ " and 1".

I.e. Slot opening is applicable also from a range of 13-35 mm for each slot based on customer requirement.

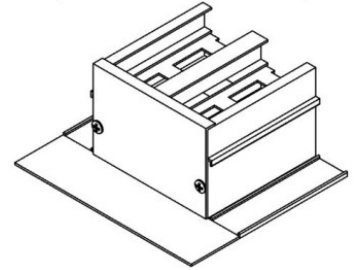
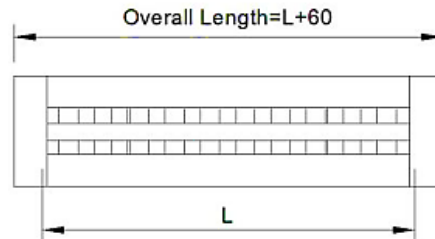
NO. OF SLOTS	S = 3/4" or 20 mm				S = 1" or 25 mm			
	N		F		N		F	
	inches	mm	inches	mm	inches	mm	inches	mm
1	1-13/16	46	3-1/8	79	2	50	3-1/4	83
2	3-1/4	83	4-5/8	117	3-9/16	91	4-15/16	125
3	4-3/4	120	6-1/16	154	5-3/16	132	6-9/16	166
4	6-1/4	158	7-1/2	191	6-7/8	174	8-1/8	207
5	7-11/16	195	9	228	8-7/16	215	9-3/4	248
6	9-1/8	232	10-7/16	265	10-1/16	256	11-3/8	289
7	10-9/16	269	11-7/8	302	11-11/16	297	13	330
8	12-1/16	306	13-3/8	339	13-5/16	338	14-5/8	371

End Cap Construction

Since linear slot diffusers are typically supplied in long, interconnected sections, they feature open ends for seamless connection. These open ends are subsequently sealed using end caps. It's crucial to understand the different end cap configurations and their respective impacts on the diffuser's overall length.

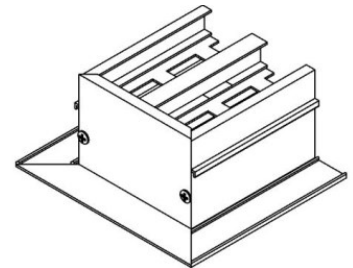
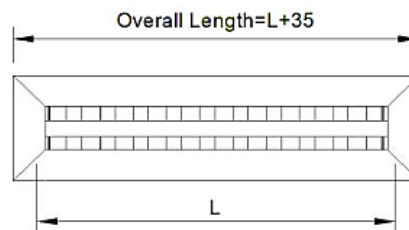
Flanged End Caps:

Its Our Standard End Caps arrangement.
Face length is increased by 60mm
(1 end cap from each side 30mm each).



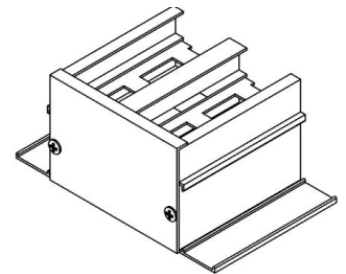
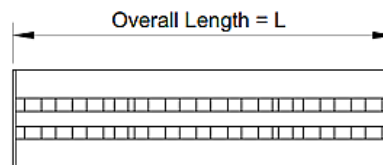
Mitered End Caps:

Face length is increased by 35mm (1 end cap from each side 17.5mm each).



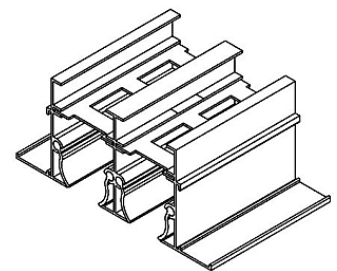
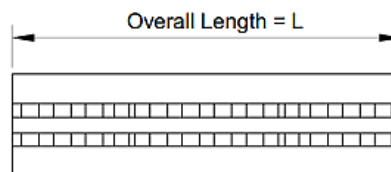
Flat End Caps:

As illustrated it has no impact on the length.



Without End Caps:

Obviously, the length is not affected

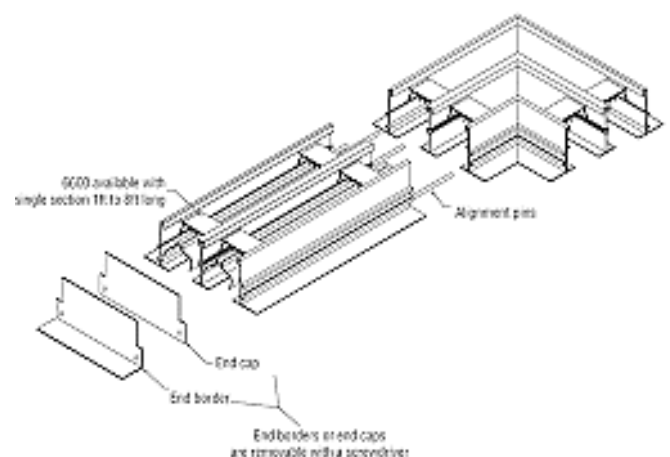


Combinations:

Combinations are used when a customer wants to attach one linear diffuser of the supply type to another of a different type (return or dummy).

In this case, there will be no end cap on one side for connection.

Similarly, if the diffuser is intended to be a middle piece, it will not have end caps, and in this case, we will consider the exact (not nominal) length.



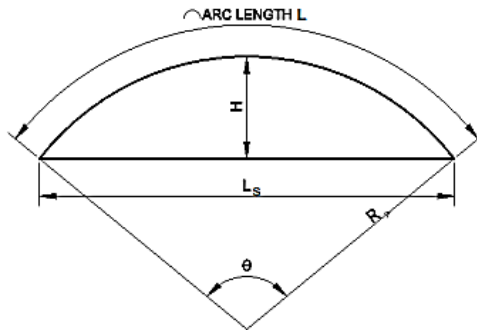
In general, during the order generation process, we should pay attention to the end cap requirements, especially the exact and nominal neck sizes, particularly in the case of combinations.

Curves:

You must know that curved diffusers are supplied without deflector and with only 1 single strip whether they are supply or return.

1. When the customer requests the diffuser to be curved we should ask him to clarify at least 2 additional dimensions ex: (L & R) in order to define the curve accurately as per design requirement, otherwise please request a forma to cut.

You have to know that the minimum radius for any curve or circle is 1m, otherwise you should consult CSD.



θ	=	Angle in degrees.
R	=	Radius.
H	=	Segment height.
	=	$R \times (1 - \cos(\theta / 2))$
L_s	=	Segment Length.
	=	$2 \times R \times (\sin(\theta / 2))$
L	=	Arc length
	=	$\theta \times (\pi / 180) \times R$

2. As Shown in Fig.1 the construction of curve differs according to Installation place (Ceiling or wall), Therefore you have to specify clearly whether the Customer need the Curved Slot diffuser to be Ceiling or wall.



Fig.1

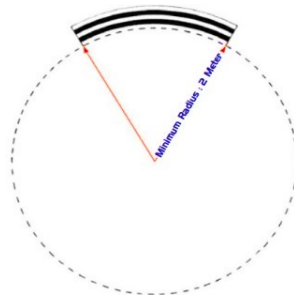
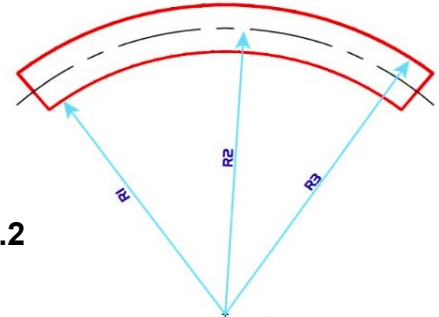


Fig.2

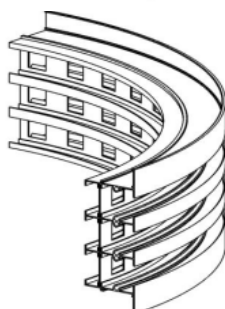
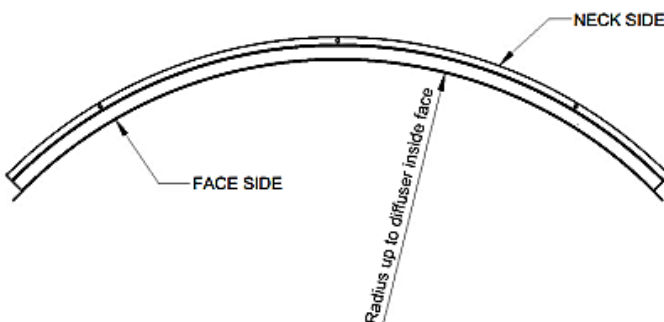


Note: * Please specify the proper radius (R1, R2 or R3) when ordering curved linear slot diffusers.

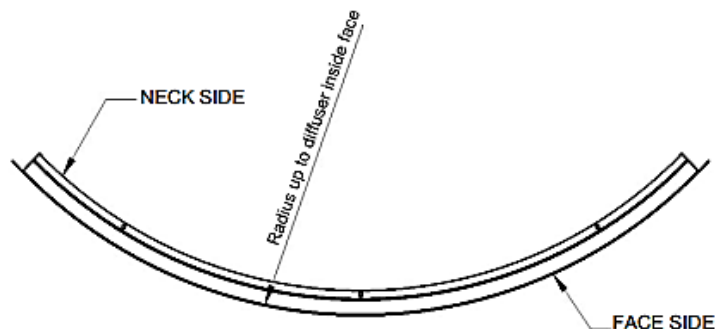
3. As Shown in Fig.2 (For Curve Ceiling), we should ask the customer to clarify clearly from where the radius is measure (R1) from the inside part of the curve or (R3) from outside the curve or (R2) from the center of the curve. (We usually order by either R1 or R3, R2 is only for illustration).

4. In case of wall curves you have to clarify whether the curve to be concave or convex by clarifying the curve to be wall inside or outside as illustrates below.

LSD-VI (INSIDE WALL)

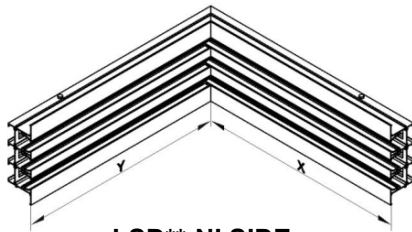


LSD-VO (OUTSIDE WALL)



Corners:

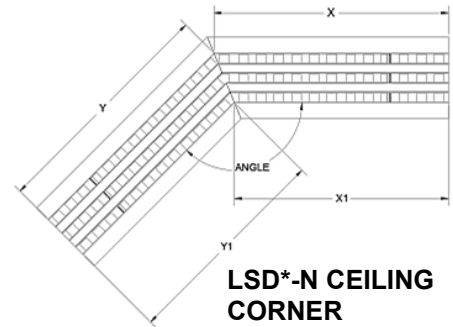
Similar to Curves You have to clarify if the Corner is Wall inside or Wall Outside or Corner Ceiling. Also for ceiling corners you should ask the customer to clarify the corner dimensions either measure from inside or outside as clarified below.



LSD-NI SIDE
WALL - INSIDE**



LSD-NO SIDE
WALL - OUTSIDE**



**LSD*-N CEILING
CORNER
(FACE VIEW)**

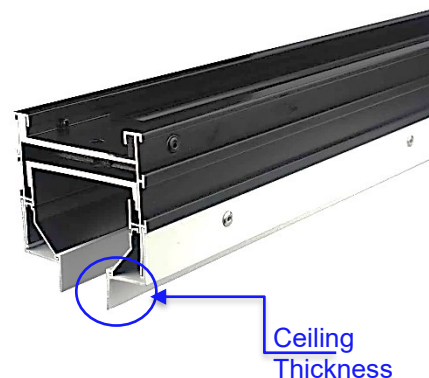
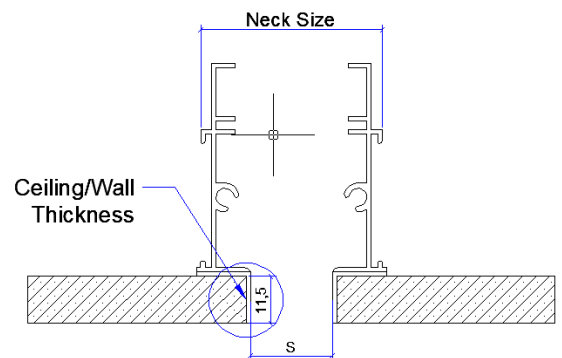
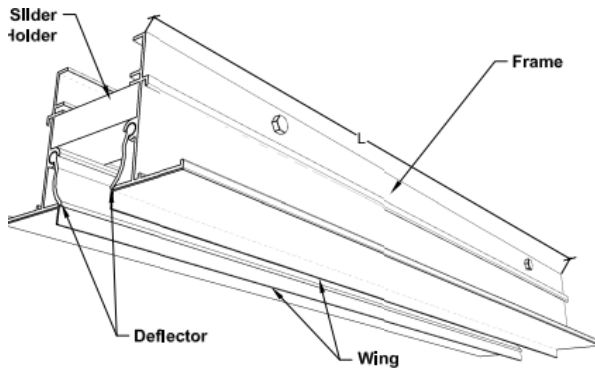
-You have to know that corners are defined by 3 parameters corner angle, X and Y (Outside) or corner angle x1 & y1 (Inside) in case of ceiling, in case of wall there is no difference in X & Y.

- CORNER PIECES - Minimum X1 or Y1 = 6"

- Available standard angles 90° and 135°.

Hidden Frame

In Case of the customer request the diffuser to be hidden type we should ask the customer to clarify the ceiling or wall thickness.



Performance:

When selecting a linear slot diffuser from the table provided (Table.1), it is important to consider the following factors:

- **Air flow rate per meter:** This is the amount of air that the diffuser can flow per meter of length. It is important to select a diffuser that has an air flow rate that is appropriate for the size of the room and the number of people in the room.
- **Face velocity:** This is the speed of the air as it exits the diffuser. It is important to select a diffuser that has a face velocity that is comfortable for the occupants of the room.
- **Throw:** This is the distance that the air will travel from the diffuser before its velocity drops to a specified level. It is important to select a diffuser that has a throw that is appropriate for the size of the room.
- **P, in mm H₂O:** This is the static pressure loss of the diffuser. It is important to select a diffuser that has a static pressure loss that is appropriate for the fan system that will be used to power it.
- **Noise criteria (NC):** This is the noise level of the diffuser. It is important to select a diffuser that has a noise level that is acceptable for the occupants of the room.

To select a linear slot diffuser from the table, follow these steps:

1. Determine the air flow rate per meter that you need. This can be done by calculating the CFM (cubic feet per minute) of air that needs to be flowed and then dividing by the length of the diffuser.
2. Determine the face velocity that you want. The recommended face velocity for linear slot diffusers is 2.61 to 3.28 m/sec.
3. Go to the suitable Table according to the suitable slot size (3/4" or 1") and match the corresponding number of Slots and cfm/ft.
4. The **number of Slots** and **cfm/ft** will intersect at 6 Points, 3 in case of vertical mounting (wall) and 3 in case of horizontal mounting (Ceiling).
5. Determine the throw that you need. This can be done by calculating the distance from the diffuser to the farthest point in the room where you want the air to reach.
6. Determine the static pressure loss that you can afford. This will depend on the fan system that you will be using to power the diffuser.
7. Determine the noise criteria (NC) that you are willing to accept. The recommended noise criteria for linear slot diffusers is 25 to 35 NC.

Once you have determined all of these factors, you can start to select a diffuser from the table. The table shows the air flow rate per meter, face velocity, throw, static pressure loss, and noise criteria for a variety of linear slot diffusers. Select a diffuser that meets all of your requirements.

LSDS

S – 3/4"

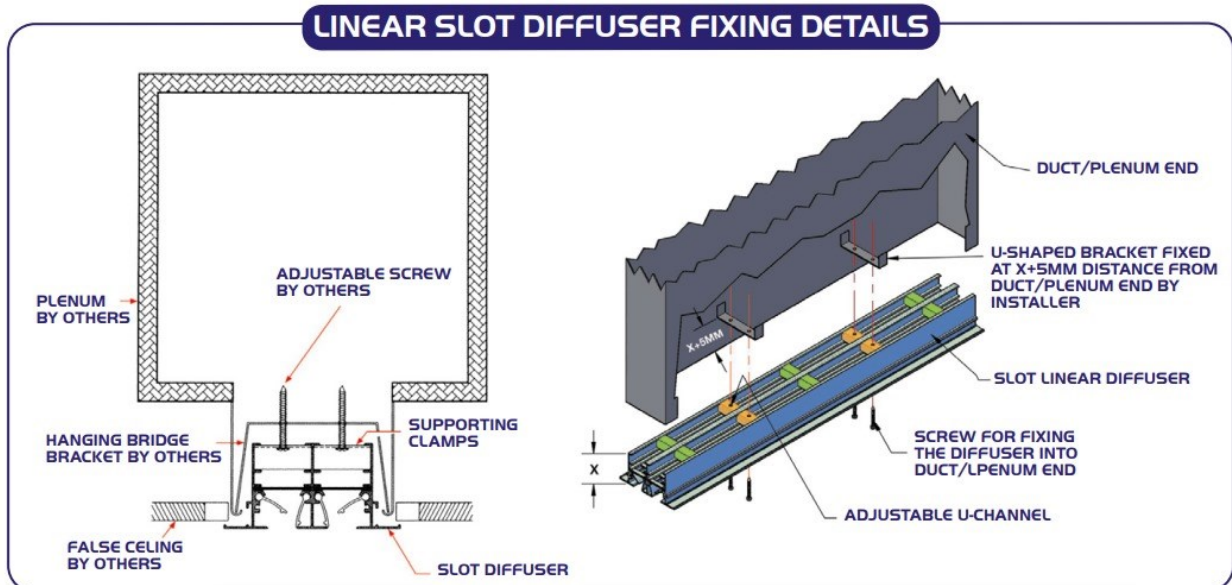
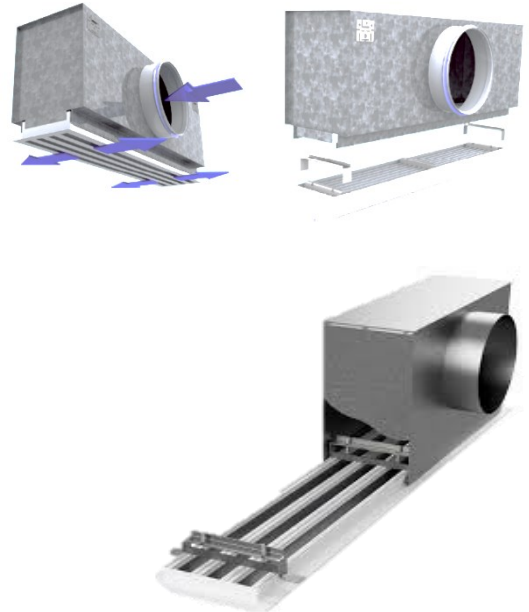
Table.1

No. of Slots	CFM/FT.	10	20	30	40	50	60	70	80	90	100	125	150	200	250	300	350	400
1	HOR. SP	.022	.089	.200	.356	.557	.802	1.092	1.426	1.804	2.228							
	THROW	7-10	13-15	15-19	18-21	20-24	22-26	23-28	25-30	27-32	28-33							
	NC	<20	<20	30	39	41	47	>50	>50	>50	>50							
	VERT. SP	.010	.039	.088	.157	.245	.353	.480	.627	.794	.980							
	THROW	2-5	9-15	16-23	22-27	25-30	27-32	29-35	31-37	33-39	35-42							
	NC	<20	<20	<20	20	27	33	38	42	46	49							
2	HOR. SP		.022	.050	.089	.139	.200	.273	.356	.451	.557	.870						
	THROW		9-13	14-19	18-21	20-24	22-26	23-28	25-30	27-32	28-33	31-37						
	NC		<20	<20	<20	28	33	38	42	46	>50	>50						
	VERT. SP		.010	.022	.039	.061	.088	.120	.157	.199	.245	.383						
	THROW		3-6	7-13	12-21	19-26	23-31	27-35	30-37	33-39	35-42	39-46						
	NC		<20	<20	<20	<20	<20	<20	23	27	30	37						
3	HOR. SP			.022	.040	.062	.089	.121	.158	.200	.248	.387	.557					
	THROW			11-16	15-21	19-24	23-26	23-28	25-30	27-32	28-33	31-37	34-41					
	NC			<20	<20	<20	22	27	31	35	38	46	50					
	VERT. SP			.010	.017	.027	.039	.053	.070	.088	.109	.170	.245					
	THROW			4-8	7-13	10-20	15-26	20-30	25-34	28-38	31-42	39-46	43-51					
	NC			<20	<20	<20	<20	<20	<20	<20	<20	26	32					

Installation

Linear slot diffusers (Supply) are commonly paired with plenum boxes to ensure consistent air distribution along the diffuser's length, maintaining optimal performance.

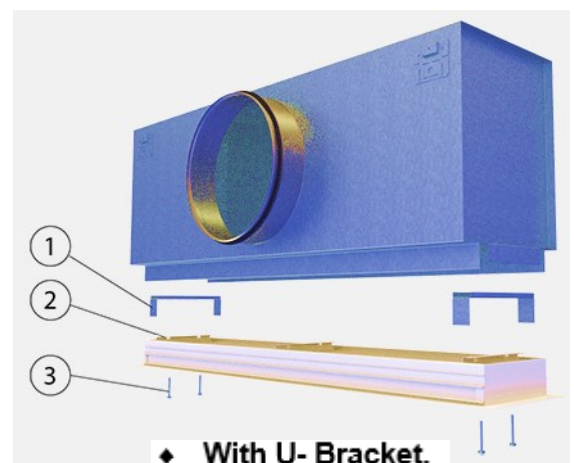
Plenum boxes are crucial components of HVAC systems, working closely with linear diffusers and grilles to achieve even airflow distribution. These boxes play a vital role in maintaining optimal performance characteristics of the diffuser system and will be explained in more detail later.

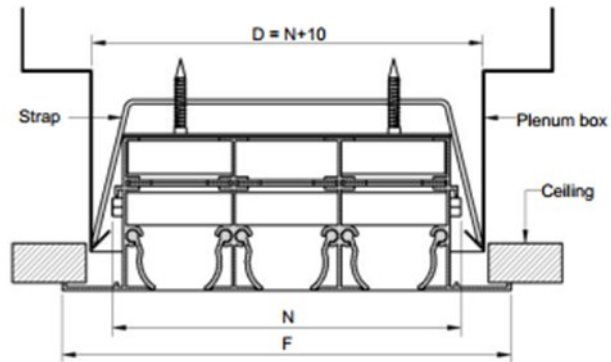


1-Installation with Plenum box.

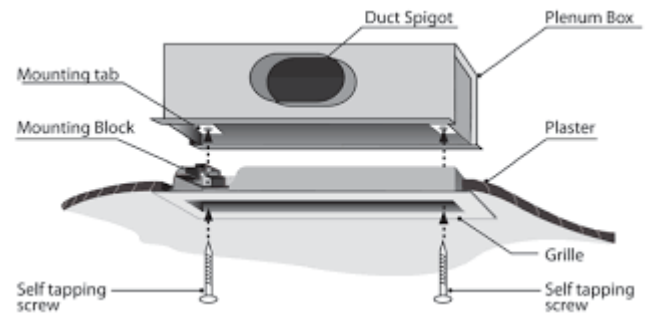
There are many ways for connection with Plenum box:

- ◆ With Clamp & Strap.
- ◆ With U- Bracket.
- ◆ Mounting bracket & Screws





♦ With Clamp & Strap.



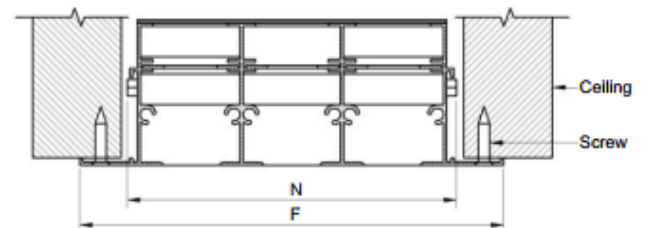
♦ Mounting bracket & Screws

For more clarification you can scan the QR code.

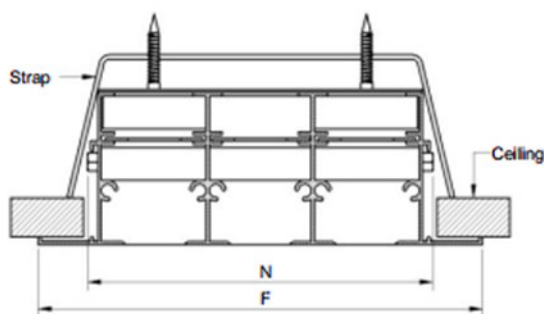


2- Installation without Plenum box.

- ♦ With Fixing Screws
- ♦ Strap and Screw.



♦ With Fixing Screws



♦ Strap and Screw.



Air Grilles & Registers

Grilles and Registers: Essential Components of HVAC Systems

Grilles and registers are crucial elements of heating, ventilation, and air conditioning (HVAC) systems, playing a pivotal role in regulating airflow and maintaining a comfortable indoor environment. While they share similarities in appearance and function, there are distinct differences between the two.



Construction and Components

Grilles and registers typically consist of three main components:

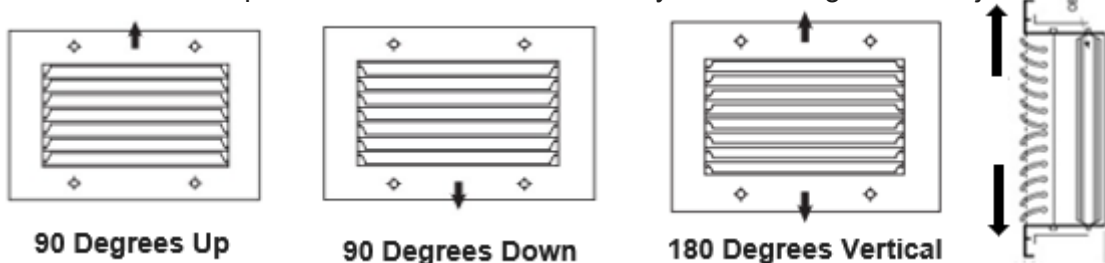
1. **Frame:** The frame provides a sturdy structure that houses the blades and damper. It is typically made from extruded aluminum (Standard) or stainless steel to withstand regular use and harsh environments. The frame is being separated from the grilles by nylon bushings. This method of assembly ensures quiet, smooth and rattle free operation.
2. **Blades:** The blades, also known as vanes, are the primary components responsible for directing airflow. They can be fixed or adjustable, depending on the desired level of control over air distribution. (Our standard Grilles and registers are manufactured from a set of Parallel blades spaced evenly with $\frac{3}{4}$ " apart).
3. **Damper:** The damper (absent in grilles) is a mechanism that regulates the airflow volume. It consists of movable blades that can be adjusted to increase or decrease the air passage area.

Types of Blades and Airflow Direction

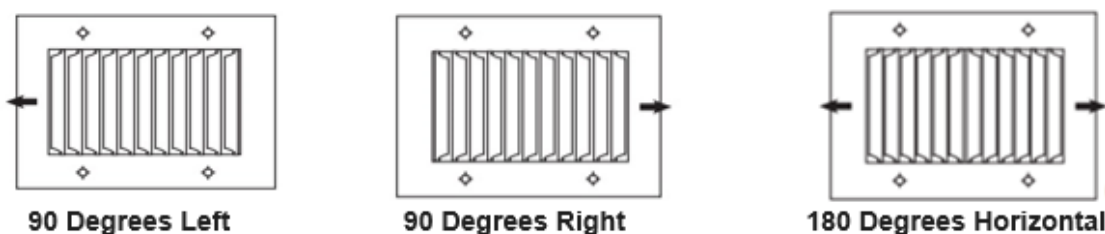
Blades in grilles and registers can be categorized based on their deflection and adjustability:

1. **Single Deflection Blades:** These blades deflect airflow in a single plane, either horizontally or vertically. They can be fixed or adjustable.

- **Horizontal Blades:** Horizontal blades allow fine-tuning the air throw angle in vertical direction either upwards or downwards with many different angles and adjustments.



- **Vertical Blades:** Vertical blades allow fine-tuning the air throw angle in a horizontal direction either left or right with many different angles and adjustments.



2. **Double Deflection Blades:** These blades provide more flexibility in airflow control, allowing deflection in two, three, or four directions. They are typically adjustable and offer a wider range of air throw patterns.

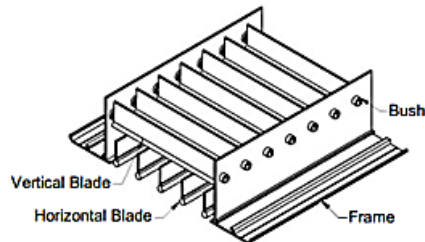


Fig. A (AGDDH) STD.

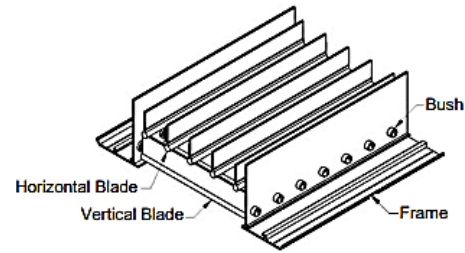


Fig. C (AGDDV)

By default, the first row has horizontal blades, and the second row has vertical blades. However, this can be reversed upon customer request.

They Also Come as Register by simply adding OBD.

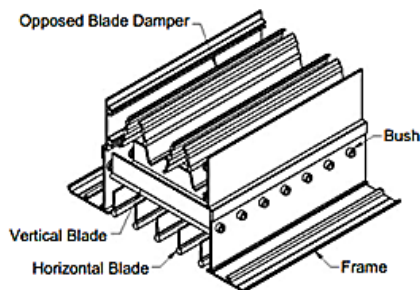


Fig. B (AGDRH)

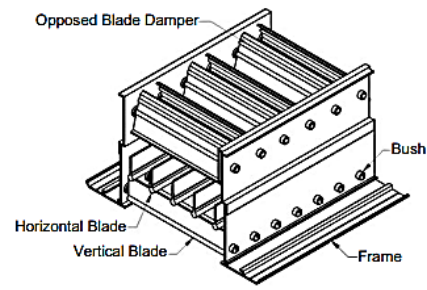


Fig. D (AGDRV)

The blades in grilles/registers can be adjusted manually or be group operated by using a lever or screw mechanism. You can Scan the QR Code for further clarification.



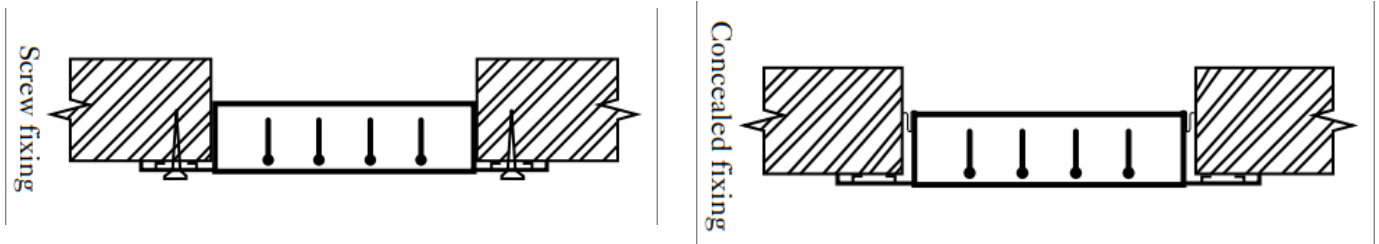
-For grille/registers with blade lengths exceeding 50cm, a blade divider is mandatory (maximum blade length is 50cm). While blade dividers are generally available at any distance, those shorter than 50cm may incur additional charges. For blade lengths greater than 50cm you have to explain to the customer that it is not applicable.

Fixed Blades vs. Adjustable Blades

Fixed blades are more commonly used in grilles, as airflow control is not typically required for return air applications. Adjustable blades, on the other hand, are essential in registers to regulate the volume and direction of supply air.

Installation

They can be installed either by fixing screws or by using concealed strips as shown:



Curved Grilles

These grilles are specifically designed for direct mounting on rigid round ducts, with a curved face.

Construction:

Frame: High quality extruded aluminum profiles with 30mm flange width as standard.

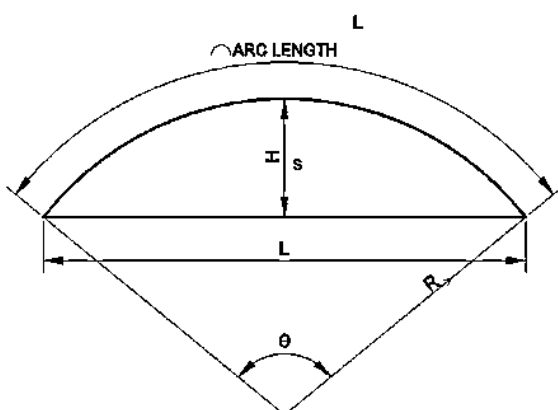
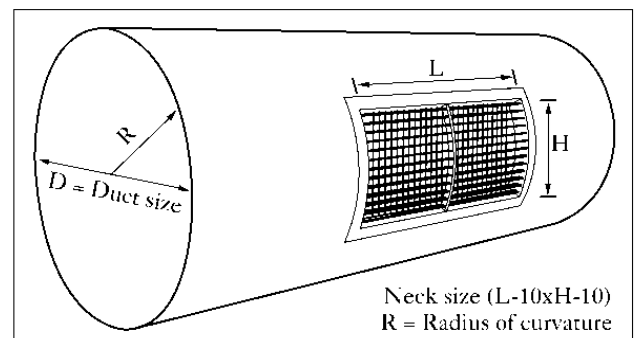
Blades: Airfoil blades from extruded Aluminum Profiles.

Blade Spacing: $\frac{3}{4}$ " (20mm) Spacing.

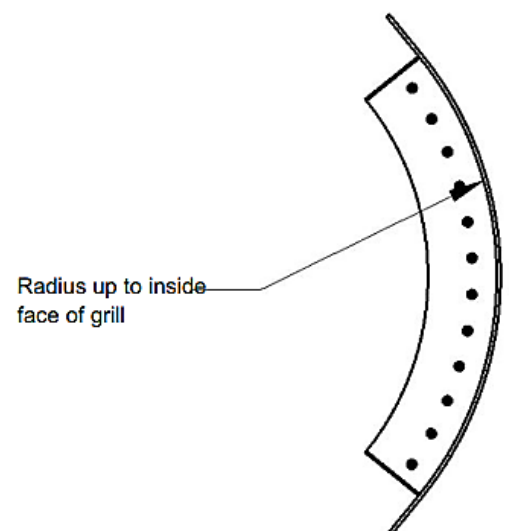
Requesting Quotation:

Curved grilles have to be identified by 3 parameters as illustrated below.

Ex: Length, Width (Nominal Neck Size) & Duct Radius.

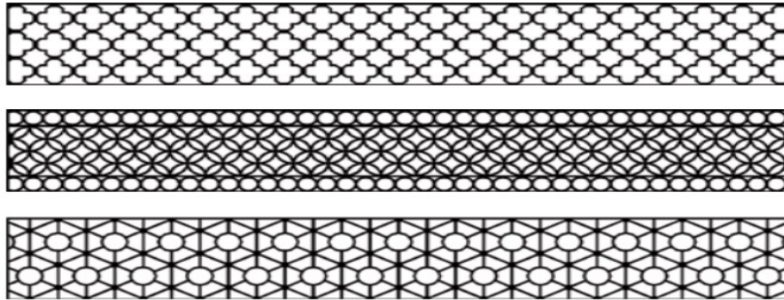


- θ = Angle.
- R = Duct Radius.
- H = Segment height.
- $= R \times (1 - \cos(\theta / 2))$
- L_s = Segment Length.
- $= 2 \times R \times (\sin(\theta / 2))$
- L = Arc length
- $= \theta \times (\pi / 180) \times R$



Decorative Air Grilles:

Air grilles are not merely functional components of HVAC systems; The customizable nature of air grilles is a key factor in their popularity among customers. The ability to tailor their appearance to specific design requirements and enhance the overall aesthetics of a space makes air grilles a valuable addition to any building.



If a customer asks for a special design for grilles, you should consult the CSD team for help to draw and design the desired pattern that meets the customer's needs.

Construction:

In case of decorative grilles their construction mainly based on 3 components

Frame: High quality extruded Aluminum profile with 1.1mm thickness and 30mm flange.

Core: Made up of Aluminum sheet, machined to the desired pattern required by customer.

Damper frame and Blades: Our Standard OBD made of high-quality extruded aluminum profiles with black matt finish.

Finishes:

- Powder Coated finish as per RAL color codes (RAL 9016 is Standard).
- Flexibility of finishing is available as an option.

Performance:

- The free area mainly depends on the design pattern of the grille.
- The amount of unobstructed airflow through a grille is primarily determined by the pattern of its design.
- Performance data is only available for our standard grilles, as decorative grilles are not intended to be evaluated for their airflow performance

Installation:

- Fixing methods are the same as our standard grilles either by screws or concealed clips.
- These decorative grilles can be fixed on ceilings or walls.

Linear Bar Grilles

Grasping the principles of linear slot diffusers and air grilles make understanding linear bar grilles straightforward. Linear bar grilles share similarities with air grilles in terms of direction and flow control, as they can be equipped with single deflection blades or double deflection blades. Additionally, they resemble linear slot diffusers in their versatility, being available in straight lines, corners, curves, and even combinations, along with diverse end cap constructions. However, there are also notable distinctions between these types of diffusers.

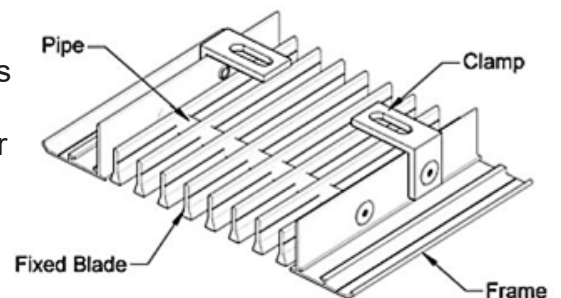


Linear Bar Grilles: Understanding Direction, Flow Control, and Construction

Linear bar grilles, as the name suggests, are air grilles featuring horizontal bars that run across the length of the grille. These bars are typically fixed (Unlike Air Grille), Similar to air grilles, linear bar grilles can be equipped with single or double deflection blades, enabling further customization of airflow patterns.

Application:

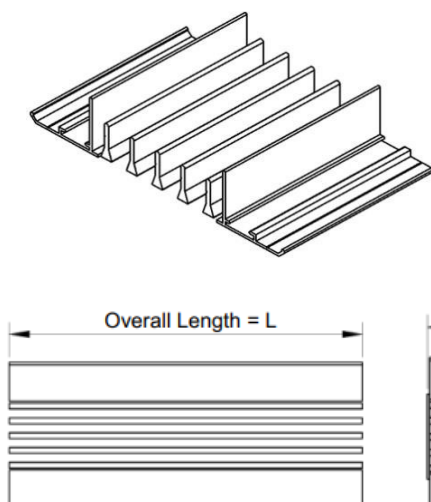
Linear bar grilles are used to distribute air in long spaces such as corridors or large areas such as open halls. They differ from linear diffusers in that they discharge air in a fixed direction, while diffusers move the air in a vortex-like pattern. As a result, this type is typically used for wall mounting.



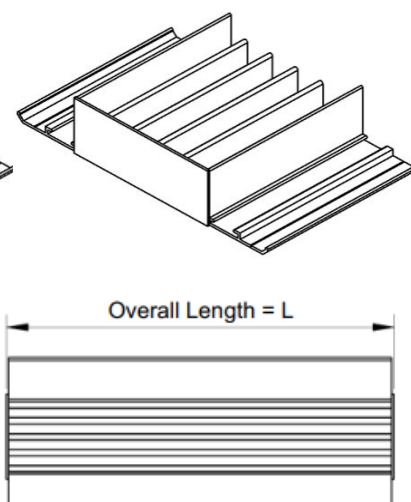
Components:

- Frame:** The frame surrounds the grille core in a symmetrical manner on all four sides. The corners of the frame are mitered (Standard) at a 45-degree angle, Similar to the linear slot diffuser, other end cap constructions are available. Unlike linear Slot diffusers the nominal neck size is Considered with Linear bar grilles making Neck and Face difference 50mm (LSD difference = 60 mm).

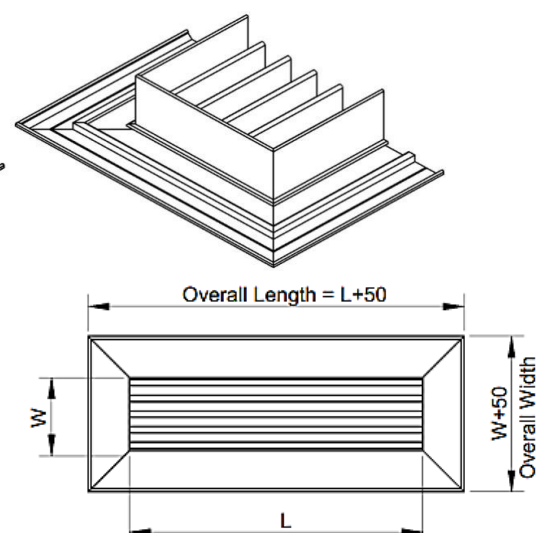
Without End Cap



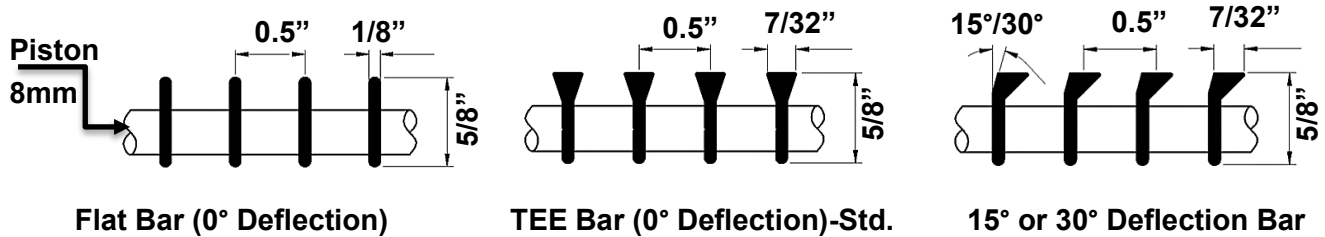
Flat End Cap



Mitered End Cap (Std.)



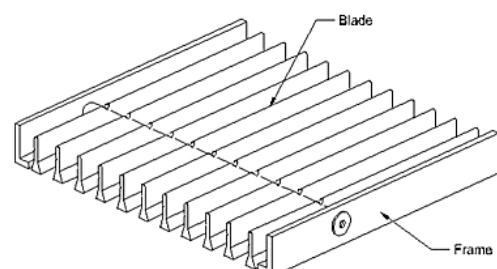
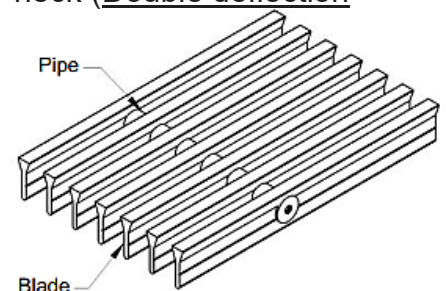
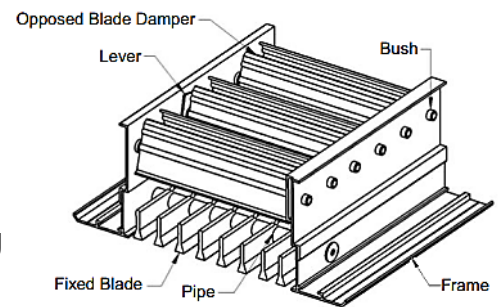
- **Bars:** The bars are fixed in a horizontal position and are connected together by an aluminum tube with an 8-mm diameter. As a result, they are not movable. The bar is in the shape of a T (Standard), so the direction of the air from either side of the bar is symmetrical (0 deflection).



- Other bar sections are available that are not in the shape of a T, such as the straight section, which also does not change the direction of the air (0 deflection), and sections that direct the air at a deflection angle of 15 or 30 degrees.
- The spacing between the bars is fixed at 0.5 inches and cannot be changed. The bars are secured using a fixed-dimension mold.
- Similar as Air Grilles they can be either with single or double blade Construction.

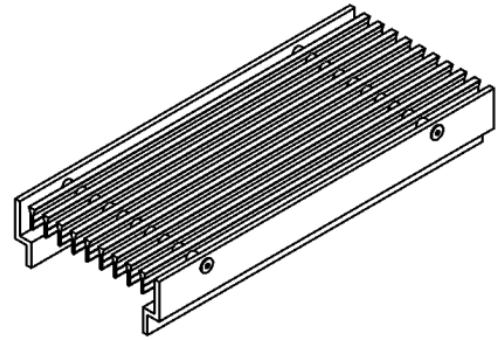
Options:

- **Supply grilles (SLBG):** A damper with opposed blades is installed. The blades are in the opposite direction of the blade that follows, which allows the damper to be opened and closed, thereby controlling the amount of air passing through.
- **Return grilles (RLBG):** Consists of a frame and bars only, without the damper.
- **Bar and blade grilles:** In this type, vertical blades are installed behind the bars. The blades are rotatable, so the direction of air discharge can be controlled. It can be with a damper, in which case another frame with a longer neck (Double deflection frame) is used to accommodate the bars and blades.
- **Frameless bar grilles (LBGC):** A core only (only Core), without a frame, as the bars, as mentioned above, are connected together and therefore are cohesive without the frame.
- **Flangeless bar grilles (LBGL):** In this type, a different frame in the shape of an L is installed, but with a back flange, not a front flange as in the standard type.



The standard frame can be installed, but with the flange cut to give a shape close to that required in this type.

- **Floor bar grilles (LBGF):** Have a relatively stronger frame and are mounted on the floor. They are designed to withstand a force of 1000 kg per square meter to withstand the passage of people.



- **Corner and curved bar grilles:**
Can be manufactured in the same shapes and sizes as linear Slot diffusers.
 - ♦ **Corner:** You have to Specify Corner Ceiling (Dimensions inside or outside) or Corner wall (Specify Wall inside or outside), Noting that minimum inner dimensions for X & Y is 6".
 - ♦ **Curve:** You have to Specify Curve Ceiling (Dimensions inside or outside) or Curve wall (Specify Wall inside or outside), Noting that minimum inner radius doesn't exceed 1m.

Selection:

Selection is made in the same way as linear diffusers, but you will find in the table that the flow rate along the length has a different range for each width of the grille, and the top row contains the speed.

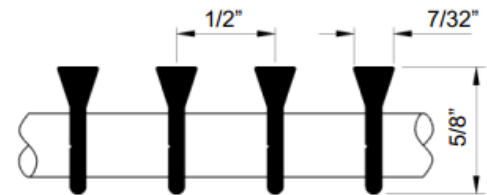
Linear Bar Grille Selection Procedure:

1. Choose the Appropriate Table by identifying:
 - ♦ Identifying the suitable blade construction (Flat bar- T-Bar, 15° or 30° deflection).
 - ♦ Determine the type (Supply or Return).
2. Calculate Velocity (FPM)

$$\text{Velocity (FPM)} = \text{Flow Rate (CFM)} / \text{Free Area (ft}^2\text{)}$$
3. Determine NC and Throw: Locate the intersection of the bar width row and the velocity column in the appropriate table.
 - ♦ Read the corresponding values for: NC, throw for different applications (Ceiling, Floor, Sidewall), Total Pressure (in.wg).

Linear Bar Grille **Supply** performance data)

1/2" spacing, 0° deflection (T bar)



NOMINAL WIDTH (Inches)	AREA Sq. Ft. (Effective free area/ Linear foot)	VELOCITY (FPM)	400	600	800	1000	1200	1400	1600	1800	2000
		TOTAL PRESSURE (in. W.G.)	0.010	0.022	0.040	0.063	0.089	0.121	0.159	0.202	0.250
2	.045	Flow, CFM / Foot	18	27	36	45	54	63	72	81	90
		Throw, Feet									
		Sill or floor	1-1	4-4	7-7	9-10	11-13	14-16	16-18	17-20	19-21
		Sidewall	5-7	9-12	11-16	14-20	16-23	19-26	21-28	22-30	25-33
		NC	<20	<20	<20	<20	23	28	32	35	38
2½	.066	Flow, CFM / Foot	26	40	53	66	79	92	106	119	132
		Throw, Feet									
		Sill or floor	2-2	6-6	8-9	12-13	14-16	17-19	20-22	22-23	23-24
		Sidewall	6-9	9-12	12-17	16-22	19-25	21-28	25-32	28-36	30-39
		NC	<20	<20	<20	20	25	30	34	37	40
3	.088	Flow, CFM / Foot	35	53	70	88	106	123	141	158	176
		Throw, Feet									
		Sill or floor	2-2	7-7	10-11	13-15	16-18	19-21	22-24	24-25	26-27
		Sidewall	7-10	11-15	14-19	17-23	21-27	24-31	27-34	31-39	34-42
		NC	<20	<20	<20	21	26	31	35	38	41

Additional Considerations:

- **Material:** Linear bar grilles are typically made of extruded aluminum, which is lightweight, durable, and corrosion-resistant. However, other materials, such as stainless steel, can be used for special applications.
- **Finish:** Linear bar grilles are available in a variety of finishes, including anodized aluminum, painted aluminum, and powder-coated aluminum. (RAL 9016 is the Standard Finish).
- **Accessories:** Linear bar grilles can be equipped with a variety of accessories, such as dampers.
- **Performance:** Linear bar grilles are known for their low pressure drop and high air volume capacity. They are also relatively quiet, making them a good choice for noise-sensitive applications.

How to order Linear Bar Grille:

- Specify Length & width (Nominal Neck Dimensions). Just In case of Straight pieces, Corners and Curves as specified.
- Specify Linear Bar Grille Type (Supply, Return, Dummy, Bar and Blade, Floor, Flangeless, Farmless).

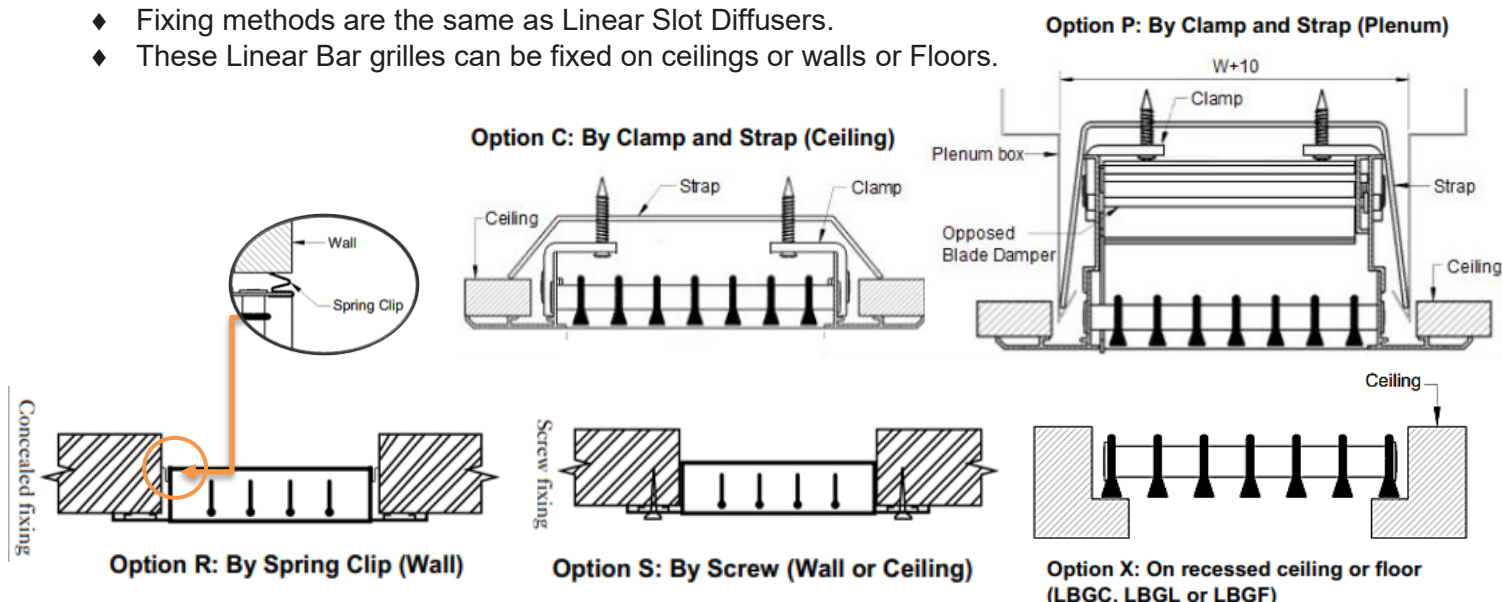
Important Remarks:

- Standard Bar is T-Bar.
- End Cap is Standard mitered type causing the face size to exceed the N.Neck size by 50mm.
- In Case of Combinations the length is Considered Exact Neck.

We rely on standard assumptions unless the customer provides alternative instructions. However, to avoid any fabrication issues that could lead to product returns, please state Clearly any customer-specified deviations.

Installation:

- ♦ Fixing methods are the same as Linear Slot Diffusers.
- ♦ These Linear Bar grilles can be fixed on ceilings or walls or Floors.



Main differences Between Linear Bar Grilles, Linear Slot diffusers and Air Grilles.

Comparison	Linear Bar Grille	Air Grille	Linear Slot
Application	Ceiling, Floor, Wall	Wall	Ceiling, Wall
Blade	Fixed (4 options for Bar)	Adjustable	No Blade Just slot openings and Deflector (Supply)
Spacing	1/2 "	3/4"	Slot opening size (13mm to 35mm)
Damper	OBD	OBD	Hit & Miss Damper
Std. End Cap	Mitered	-	Flanged.
Standard Size	Nominal Neck	Nominal Neck	Exact Neck

Other Grilles

Understanding the mechanics and construction of air grilles unlocks a world of design possibilities. Air grilles are generally composed of a set of blades and a frame. Modifying either the core construction (blades) or the face construction we can achieve endless variations of Grilles Designs.

- Here are few examples:

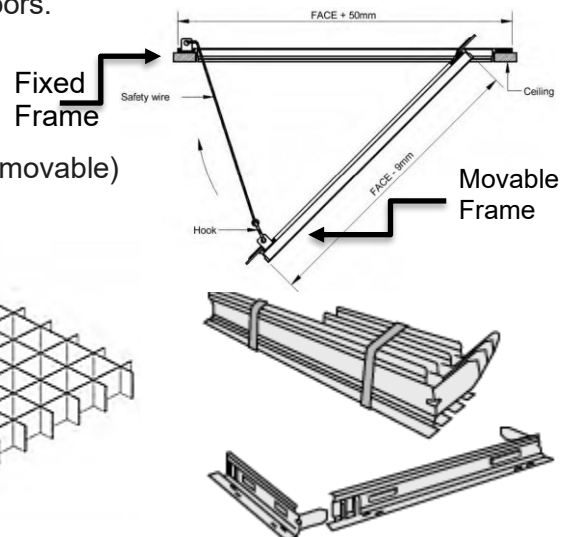
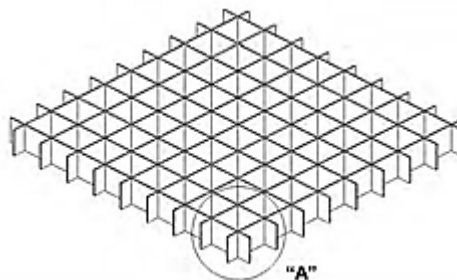
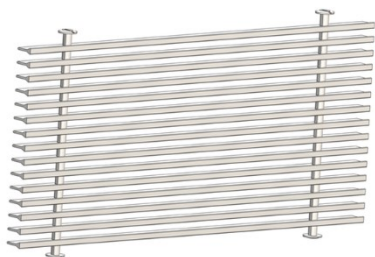


Blade Variations:

- **AG Single Deflection:** Vertical or Horizontal Blades direct airflow in one direction, providing focused distribution.
- **AG Double Deflection:** Two sets of blades offer precise control over airflow direction and spread (Face blades are Horizontal & Rear blades are vertical or vice versa).
- **AG with Adjustable Blades:** Blades are Mounted on a Nylon Bushing Allowing manual adjustment for personalized air distribution.
- **AG with Fixed Blades:** Blades fixed to the frame offering a consistent and predictable airflow pattern.
- **Customized Grille:** Customized sheet metal can be used for decorative purposes instead of Blades.
- **Egg Crate Grille:** Removed blades and added a grid design for air diffusion and visual aesthetics
- **Door Grille (Non-Vision Grille):** Inverted V-shaped blades create a visually unobtrusive barrier while allowing airflow, with double frame to be fixed on doors.

Face Construction Variations:

- **Access Door Grille:** Double frame (one fixed and one removable) with hinges allows for easy access behind the grille.



Remember:

Register: Any grille can be converted into a register by adding an opposed blade damper, enabling precise control over airflow volume.

These are just a few examples, and the possibilities are truly endless. By understanding the fundamentals of air grille design, we can create custom solutions that are both functional and aesthetically pleasing.

Door Grilles

- Door grilles are used to transfer air from one side of the door to the other so that it cannot be seen through them and so that light is not transmitted either. They give a symmetrical shape on both sides using a frame for each side.
- These grilles are also used to draw or expel air inside and outside the building, as they have a suitable design to prevent the entry of rainwater or the transmission of light, as mentioned above.

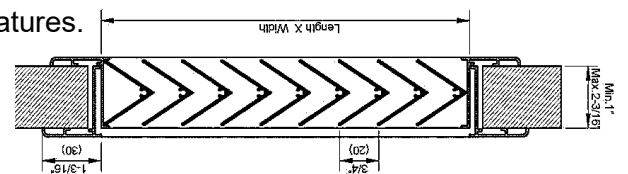


Applications:

- Maintaining air pressure balance between different areas.
- Providing ventilation in enclosed spaces without ductwork.
- Transferring air between rooms with different temperatures.

Components:

- Frames:**
 - Featuring two frames – a main frame connected to the Vanes (blades) and a Secondary frame (Without Vanes) mounted on the opposite side of the door – this design ensures a seamless visual experience from both sides.
 - The shape of the frame is similar to its counterpart (Secondary frame) used with registers. It is designed to be tight to fix the vanes, with a depth suitable for the common door depth.
- Vaness:** They are characterized by a reversed "V" shape, which causes isolation of vision and light between the two sides. The distance between the vane and the one that follows it is (3/4"-20mm) fixed, as it is fixed using a fixed-dimension mold.
- Materials:** The frames and vanes are made of cast aluminum.



Options:

- If the grille is ordered for the purpose of transferring air from and to the building, it is not necessary to use the secondary frame. In this case, an insect screen is installed on the back of the grille.

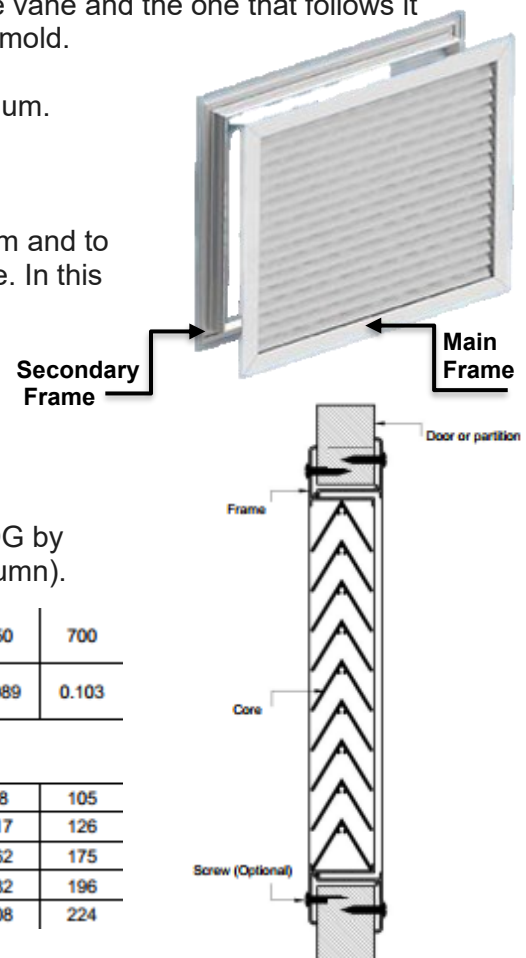
Installation:

- Fixed by Screws.

Performance:

- You Can detect the equivalent cfm for each size & ΔP of DG by simply intersecting the size (row) with the Velocity (Column).

Listed sizes (inches)	Core Velocity (fpm)	300	400	450	500	550	600	650	700
	Static Pressure (in. w.g.)	0.021	0.035	0.044	0.053	0.065	0.076	0.089	0.103
	Area Factor (Sq. ft.)	AIR CAPACITY (cfm)							
10 x 6	0.15	45	60	68	75	82	90	98	105
12 x 6	0.18	54	72	81	90	99	108	117	126
16 x 6	0.25	75	100	112	125	137	150	162	175
18 x 6	0.28	84	112	126	140	154	168	182	196
20 x 6	0.32	96	128	144	160	176	192	208	224



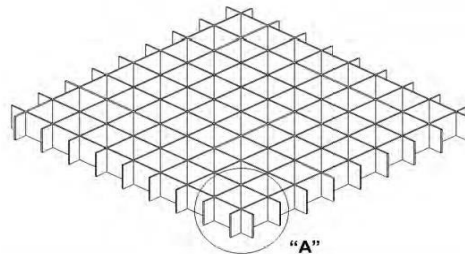
Egg Crate Grilles (ECG)

This type is used mainly if the free area through which air passes is required to be maximized. The free area represents about 90% of the total opening for air passage. Because it lacks the ability to direct air, it is rarely used for supply.

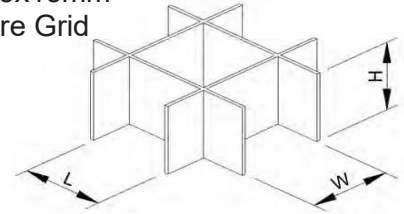
Application: Generator rooms.

Components:

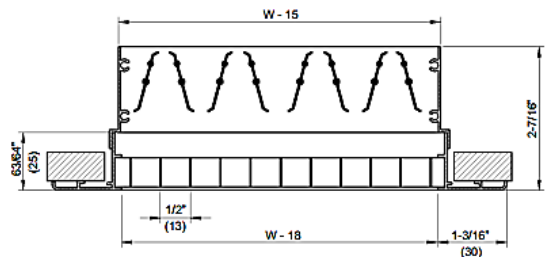
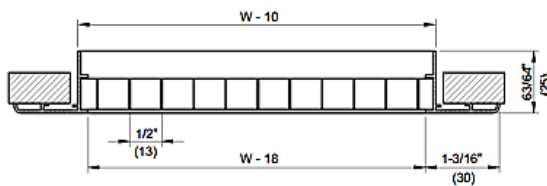
- **Frame:** The frame is in the shape of an L and is designed to reinforce the fixing of the core.
- **Core:** Consisting of linear bars that are machine cut and notched, then bonded together to form a horizontal and vertical grid structure in the shape of an Egg Crate box, and it is a matrix of square openings.



13x13x13mm
Core Grid

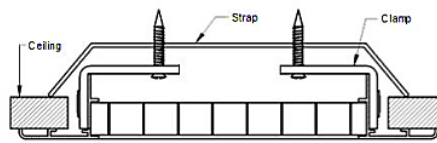
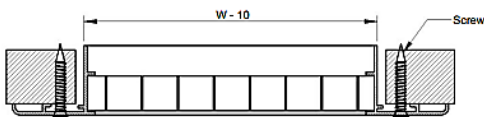


- **OBD Damper:** These grilles can be equipped with an OBD damper and are used in this case for exhaust or Supply air.

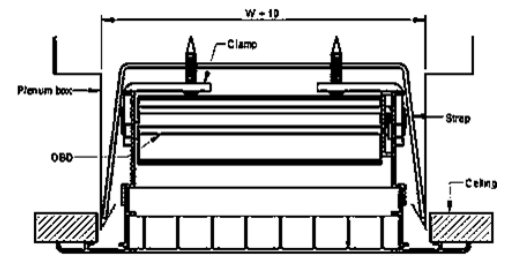


Installation:

Option S: By Screw (Wall or Ceiling)



Option C: By Clamp (Ceiling)



Option P: By Clamp and Strap (Plenum)

Performance:

- You Can detect the NC & ΔP of ECG by simply intersecting the size in inches (row) with the Velocity in FPM (Column).

ECG & ECR

Neck Size (inches)	Core Area (Sq. Ft.)	Core Velocity (fpm)	300	400	500	600	700	800	1000	1200	1400
		Velocity Pressure (in. w.g.)	.006	.010	.016	.022	.031	.040	.062	.090	.122
		Negative SP (in w.g.)	.016	.026	.042	.059	.082	.106	.165	.238	.323
24 x 20	2.971	CFM	891	1188	1485	1783	2078	2377	2971	3565	4159
		NC	<20	22	24	26	27	29	31	32	34
28 x 20	3.487	CFM	1046	1395	1743	2092	2441	2790	3487	4184	4882
		NC	<20	21	23	25	27	28	30	32	33
32 x 20	4.004	CFM	1201	1602	2002	2402	2803	3203	4004	4805	5605
		NC	<20	21	23	25	27	28	30	32	33
36 x 20	4.510	CFM	1353	1804	2255	2706	3157	3608	4510	5412	6314
		NC	<20	21	23	25	26	28	30	32	33
40 x 20	5.027	CFM	1508	2011	2513	3016	3519	4022	5027	6032	7038
		NC	<20	21	23	24	26	27	29	31	33
48 x 20	6.060	CFM	1818	2424	3030	3636	4242	4848	6060	7272	8484
		NC	<20	20	22	24	26	27	29	31	33
24 x 24	3.595	CFM	1078	1438	1797	2157	2516	2876	3595	4314	5033
		NC	<20	21	24	25	27	28	30	32	34

Access Door Grilles

These grilles are used as access ports to the inside of the false ceiling while maintaining the architectural shape. They consist of a frame fixed to the ceiling and another removable frame that contains the core, which represents the shape of the grille.

• Components:

- **Fixed Frame:** An L-shaped frame that is fixed to the false ceiling.
- **Removable Frame:** It is placed on top of the fixed frame without being fixed so that it can be removed by lifting it up, rotating it, and then removing it down again. The core is fixed to it and is connected to the fixed frame with a flexible wire to prevent it from falling suddenly.
- **Materials:** The frames are made of cast aluminum.
- **Core:** The core takes several shapes depending on the request. It can be of the type of fixed blades, egg crate, perforated, solid or Open.



AGF (Fixed Blade Core)



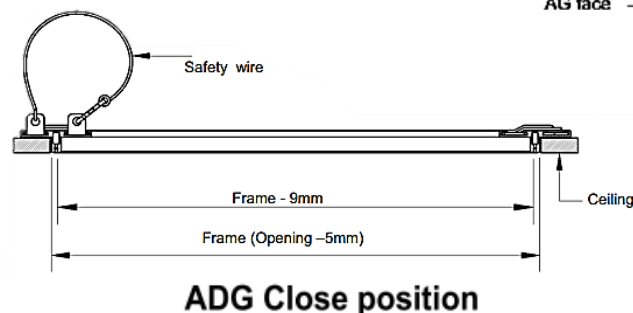
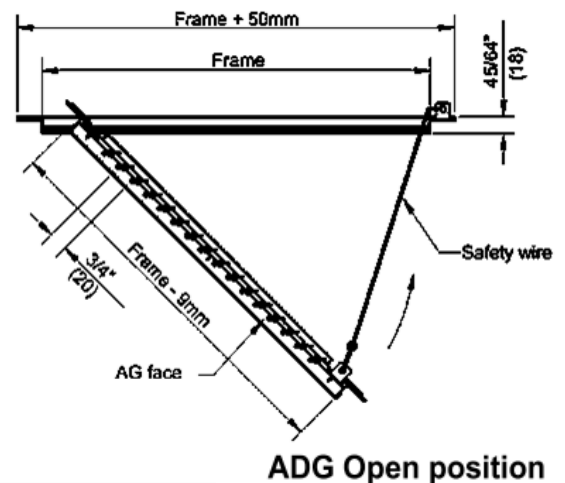
AGE (Egg Crate Core)



AGP (Perforated Core)



AGC (Open Core)



- Options:

- These grilles are not suitable for vertical installation therefore they are not suitable for wall use. However, if the customer requests access doors for this purpose, they are made differently.

- Performance:

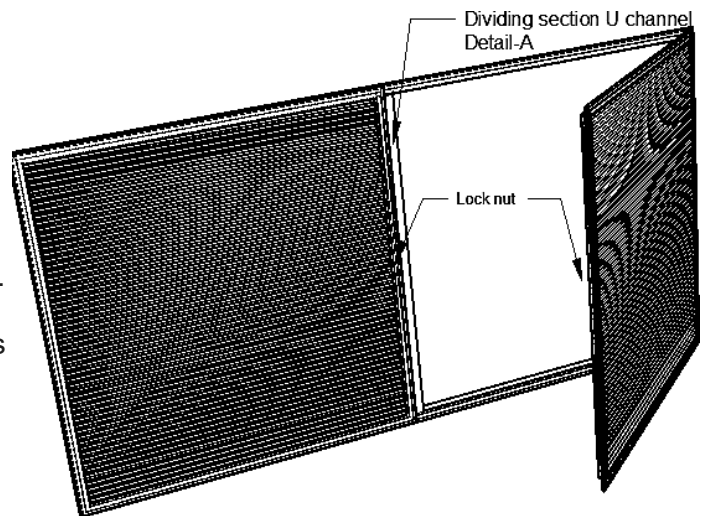
- Same as mentioned before by intersecting the velocity column (FPM) with the Size row (inch).

ADE

Listed Sizes (inches)	Area Factor (Sq. ft.)	Core Velocity (fpm)							
		200	300	400	500	600	700	800	900
		Static Pressure Drop (in. w.g.)							
		0.010	0.016	0.026	0.042	0.059	0.082	0.106	0.130
10 x 6	0.312		93	125	156	187	218	250	
10 x 8	0.441		132	176	220	264	309	353	
12 x 6	0.387		116	155	194	232	271	310	
12 x 8	0.527		158	211	264	316	369	422	474
12 x 12	0.829			332	414	497	580	663	746
18 x 6	0.592		178	237	296	355	414	474	533
18 x 12	1.281	256	384	512	640				
18 x 18	1.970	394	591	788	985	1182	1379		
24 x 12	1.733			693	866	1039	1213	1386	1560

Wall-mounted access doors:

- Usage:** These doors are used to access the inside of the false ceiling from the wall. They consist of a frame fixed to the wall and another removable frame that contains the core.



- Components:

- Fixed Frame:** An L-shaped frame that is fixed to the wall.
- Removable Frame:** It is placed on top of the fixed frame fixed to it by hinges so it can move in a pattern of 180°. The core is fixed to it and there is a cam lock nut for closure.
- Materials:** The frames are made of cast aluminum.
- Core:** The core takes several shapes depending on the request. It can be of the type of fixed blades, egg crate, perforated, or solid.
- Size:** The size of the grille is determined by the size of the access required.
- Frame Material:** The frame can be made of cast aluminum or steel.
- Core Material:** The core can be made of aluminum, steel, or other materials.

Ceiling Diffusers:

Ceiling diffusers are a vital component of air distribution systems in both commercial and residential buildings. They play a crucial role in delivering conditioned air throughout a space, ensuring optimal comfort and indoor air quality.

Benefits:

- **Efficient air distribution:** Provides optimal comfort and indoor air quality.
- **Variety of types and options:** Adapts to specific needs and aesthetics.
- **Easy installation and maintenance:** Reduces downtime and cost.

Here's a comprehensive overview of ceiling diffusers:

Types:

- **Rectangular/Square:** These are the most common type, offering balanced airflow distribution in all directions with different flow pattern 1,2,3 or 4 way. They are ideal for general ventilation and have a sleek architectural appearance.
- **Round:** Similar to rectangular diffusers, round diffusers distribute air evenly in all directions. They are a good choice for aesthetics and compatibility with flexible ductwork.
- **Jet:** Designed to discharge air at high velocity, reaching longer distances. They are commonly used in large spaces requiring efficient ventilation specially in the absence of false ceiling where the ceiling is glass ex: Air Ports or Mosques.
- **Disc Valve:** Primarily used for exhaust in bathrooms and small spaces, offering simple installation and maintenance.



Components:

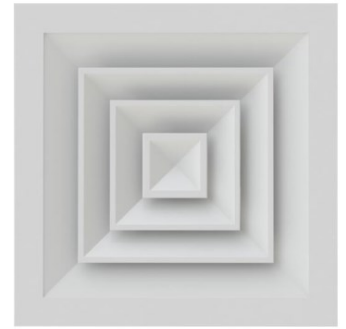
- **Frame:** Provides structural support and houses the core.
- **Core:** Directs the air through various configurations of rings, tubes, or discs depending on the diffuser type.

Options:

- **Opposed damper:** This damper is used to control the airflow rate. It is operated by levers and is typically mounted on a plenum. (rectangular/Square diffusers).
- **Butterfly damper:** Controls airflow volume (Round Ceiling diffusers).
- **Rotary Blade damper:** Offers a wide range of airflow control with a simple rotation mechanism (Round Ceiling diffusers).
- **Perforated panel:** Attached to the diffuser face, provides sound attenuation (rectangular/Square diffusers).
- **Blank panel:** Attached to the diffuser face, creates a uniform and comfortable air distribution pattern (Plaque Diffuser).

Rectangular/Square Ceiling Diffusers

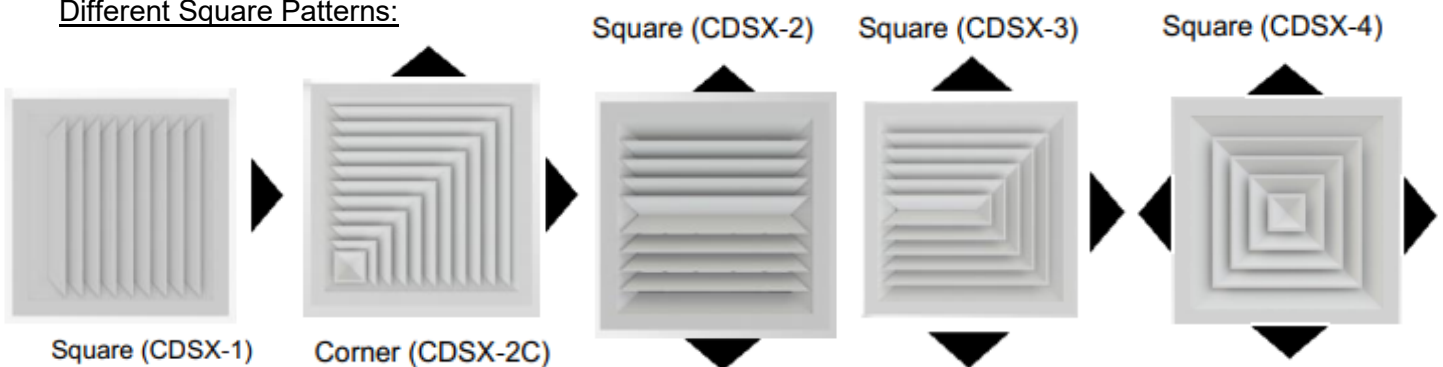
They are the most common type of air outlet. They are used to distribute air in all directions and have a balanced architectural appearance. They are also easy to handle and install, which makes them a good choice for commercial and residential applications.



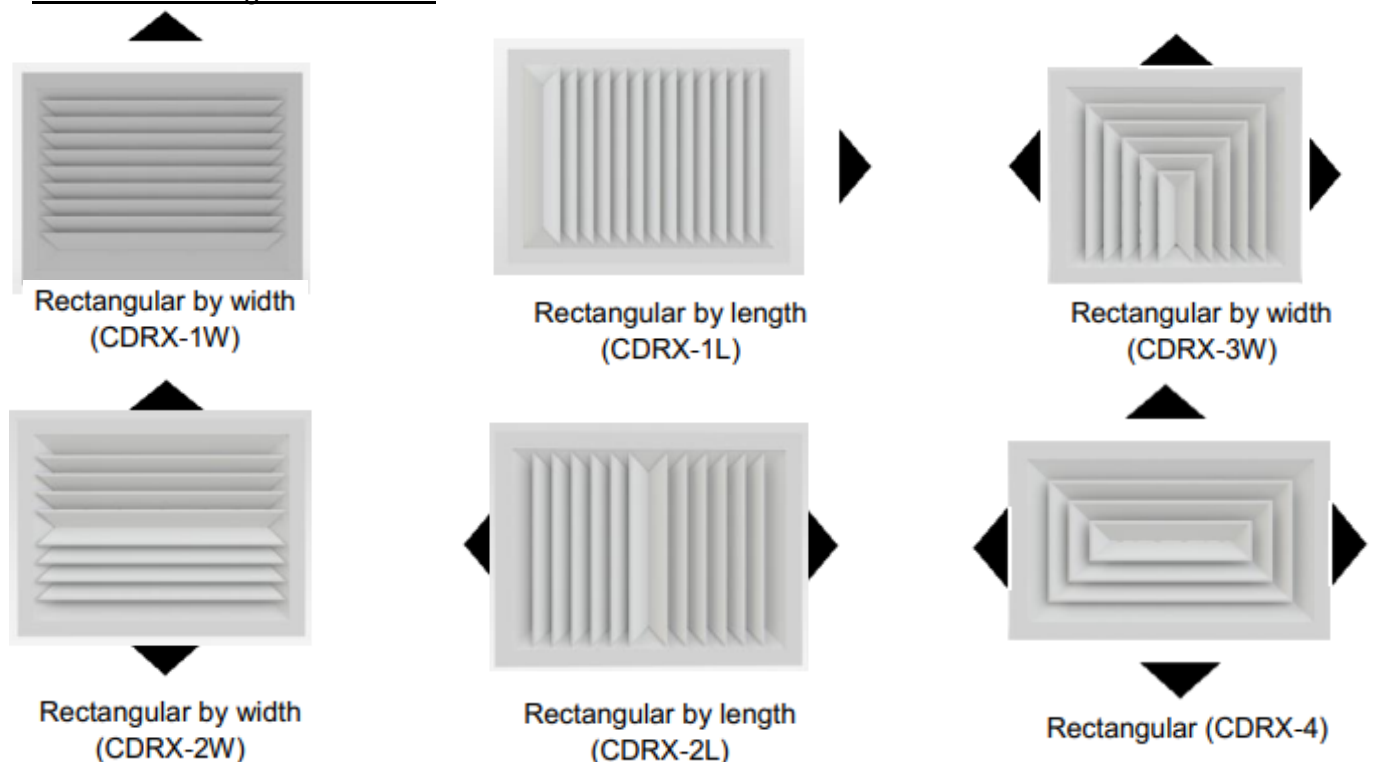
Components:

- **Core:** The core is a series of concentric Rectangular/square rings that help to distribute air evenly in all directions according to the required pattern. These rings can be arranged in different patterns to provide different air distribution patterns. (Core is Removable)
- **Flower:** The stamped center Core Piece.
- **Frame:** The frame surrounds the core and is made of cast aluminum. The difference between the face and neck sizes is always 140 mm.
- **T-Bar Frame:** A square panel Frame attached to the face to provide the desired face size when the customer demands the face size to exceeds the Neck Size by 140mm.
- **Material:** Core & Frame (Extruded AL.), Flower: Pressed Al Sheet, Additional panels: AL sheets
Standard SCD are Aluminum but they are available in Stainless steel upon request

Different Square Patterns:



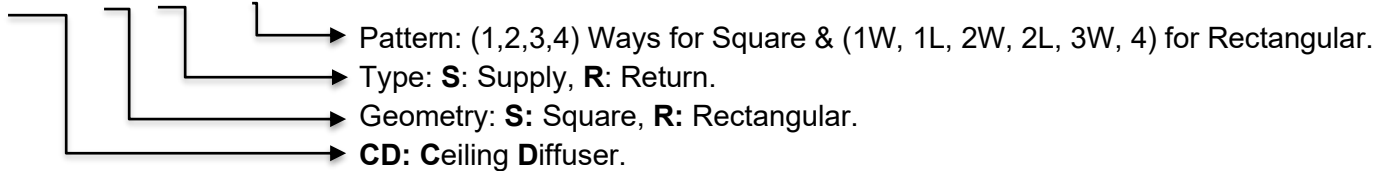
Different Rectangular Patterns:



Naming System:

Understanding how the naming system works will make it easier for you rather than memorizing the name for each type.

CDYX-N



Performance

1. Pick the Right Table:

- ◆ Square or Rectangular
- ◆ Supply or Return
- ◆ Pattern

2. Find NC & Throw (ft.) by matching velocity (FPM) and diffuser size (Inch).

3. Calculate Static pressure Drop: $\Delta P(\text{in. wg}) = \text{Total Pressure} - \text{Velocity Pressure}$

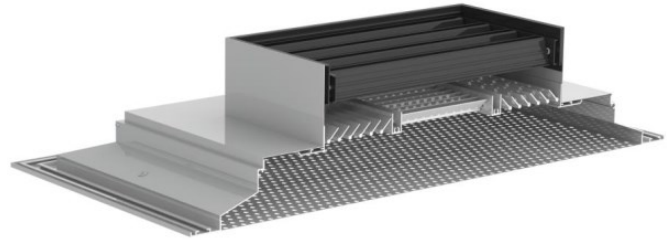
Ceiling Diffuser **Square Supply**

CDSS-4 (Performance Data)

CDSS-4 (FOUR WAY)

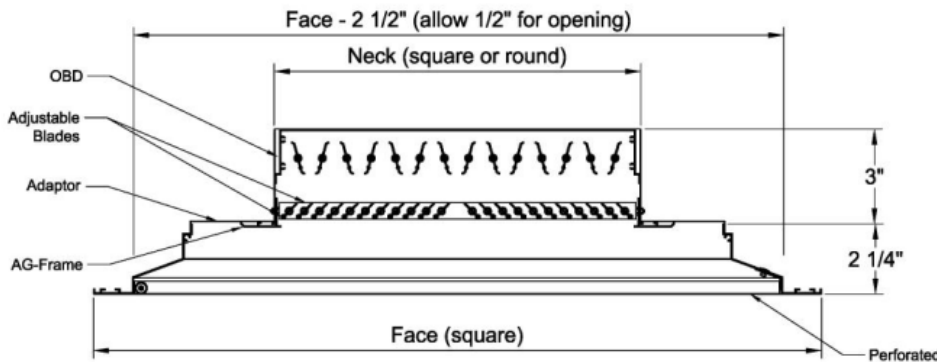


Size (inch)	Area Factor (AK) Neck Area (sq.ft.)	Neck Velocity (fpm)	200	250	300	350	400	500	600	700	800	900
		Velocity Pressure (in. w.g.)	.003	.004	.006	.008	.010	.016	.023	.031	.040	.051
		Total Pressure (in. w.g.)	.011	.017	.024	.032	.042	.066	.096	.130	.170	.245
6 x 6	Ak = .085 0.250	CFM	50	63	75	88	100	125	150	175	200	225
		THROW	2-4	3-6	4-7	5-8	6-9	7-11	8-12	10-14	11-16	13-18
		NC	<20	<20	<20	<20	<20	<20	23	27	31	35
9 x 9	Ak = .177 0.562	CFM	110	140	170	195	225	280	335	395	450	505
		THROW	4-6	5-7	6-9	7-10	8-11	10-13	11-16	13-19	15-20	17-21
		NC	<20	<20	<20	<20	<20	21	27	31	35	39
12 x 12	Ak = .306 1.000	CFM	200	250	300	350	400	500	600	700	800	900
		THROW	5-7	6-9	7-11	8-12	10-14	12-17	14-20	17-21	18-23	20-24
		NC	<20	<20	<20	<20	<20	23	30	34	38	42
15 x 15	Ak = .472 1.56	CFM	310	390	470	545	625	780	940	1090	1250	1405
		THROW	6-9	7-11	9-13	10-15	12-17	14-19	17-21	18-22	20-24	21-25
		NC	<20	<20	<20	<20	21	26	32	36	40	44
18 x 18	Ak = .674 2.25	CFM	450	560	675	785	900	1125	1350	1575	1800	2025
		THROW	7-10	8-12	10-14	12-16	13-18	16-20	18-22	20-24	21-26	22-27
		NC	<20	<20	<20	21	23	28	34	38	42	45
21 x 21	Ak = .914 3.06	CFM	610	765	920	1070	1225	1530	1835	2140	2450	2755
		THROW	7-11	9-13	11-16	13-18	15-19	18-22	20-24	21-25	23-27	24-29
		NC	<20	<20	<20	22	23	30	36	40	44	49
24 x 24	Ak = 1.22 4.00	CFM	800	1000	1200	1400	1600	2000	2400	2800	3200	3600
		THROW	8-12	10-15	12-18	14-19	16-20	19-23	20-25	22-26	23-28	25-30
		NC	<20	<20	20	23	24	31	37	41	45	50



Perforated Diffusers (CDP)

The perforated ceiling diffuser enhances the regular square diffuser by adding a perforated panel for sound absorption with 50% free area. This panel attaches to the frame with a special core arrangement that controls airflow in (1,2,3 and 4) directions and opens with hinges for easy access.



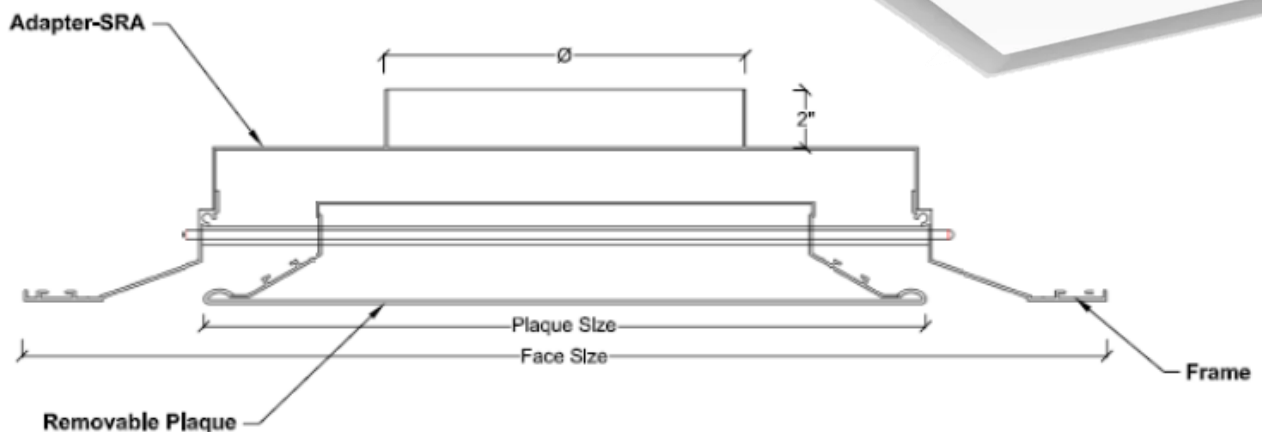
CDPR: Return

CDPS: Supply with Adjustable blades core & OBD

Plaque Diffusers (RPD)

Plaque diffusers are a popular choice for modern air distribution systems due to their sleek, minimalist design. They typically feature a flush-mounted square or rectangular face that seamlessly blends into the ceiling. It discharges air in four-way, horizontal pattern, without excessive noise or pressure drop. It is an excellent selection for variable air volume systems.

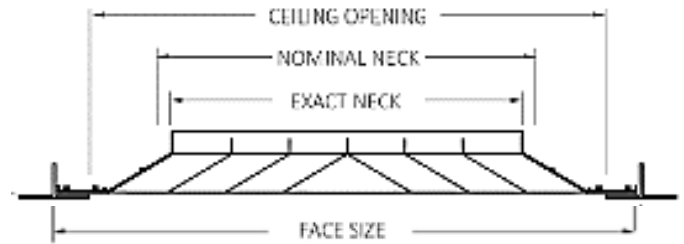
It discharges air in four-way, horizontal pattern, without excessive noise or pressure drop. It is an excellent selection for variable air volume systems.



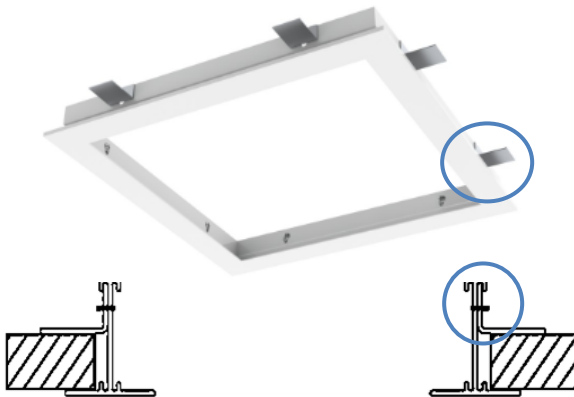
Installation:

There are various installation methods for ceiling diffusers depending on site conditions:

- ◆ Ceiling Tiles: Lay In
- ◆ Solid Gypsum board: By Using Plaster Frame
- ◆ With Plenum box
- ◆ Square/Rectangular duct
- ◆ Round Duct



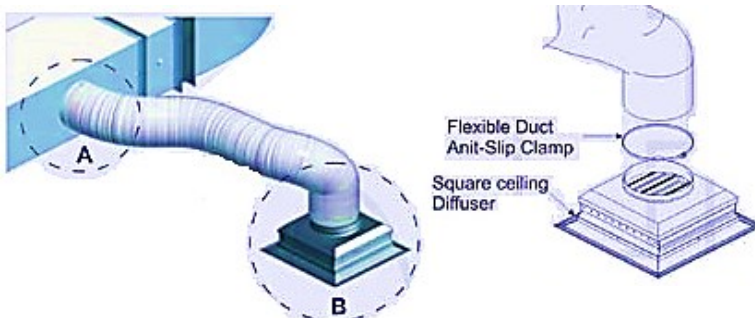
LAY IN Method-Suitable for Ceiling Tiles with Face size 60x60cm



- ◆ By Help of T-Frame (Supplied by Other) For Solid Gypsum board

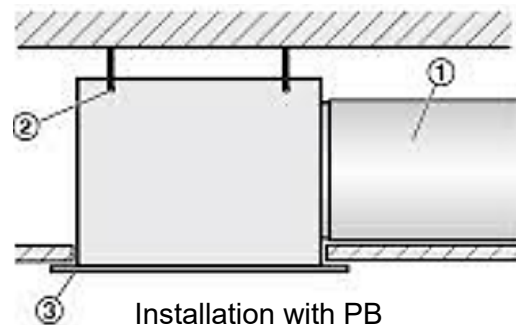


- ◆ Fixing by Screws For Solid Gypsum board

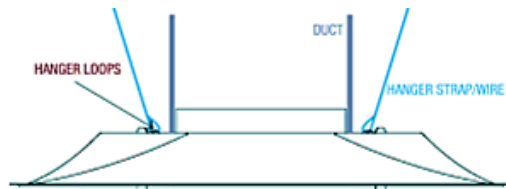


- For Flexible Duct Connection

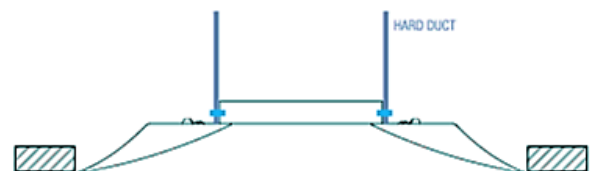
Option: Galvanized steel neck adapter square to round can be provided upon request for direct connection with Round Flexible ducts



Installation with PB



Hangers Loop Installation with Duct



Screwing Diffuser Neck To the Duct

- You Can Scan the QR Codes For more Clarification.



- ◆ Plaster Frame



- ◆ Lay-In



- ◆ Plaque Diffuser

Round Ceiling Diffusers

Round ceiling diffusers are another common type of air outlet. They provide uniform air distribution in all directions and are ideal for use with flexible ductwork.



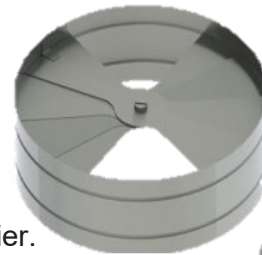
Components:

- **Frame:** The frame is made of cast aluminum or spun aluminum. For this diffuser, face size doesn't follow a set pattern like SCD. Check the catalogue to determine matching face sizes.
- **Core:** The core is a series of concentric circular rings that help to distribute air evenly in all directions. The rings can be arranged in different patterns to provide different air distribution patterns.

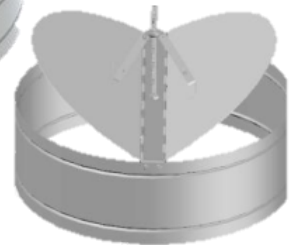
Control:

- **Rotary Blade Damper (RBD):** Several blades stacked together and secured at a central point allowing them to rotate for precise airflow Control, ideal for balancing but bulkier.
- **Butter Fly Damper (BFD):** Two hinged flat plates, operating like a butterfly valve, modulate airflow with their Swinging motion (opening and closing).
(Standard damper Used)

RBD



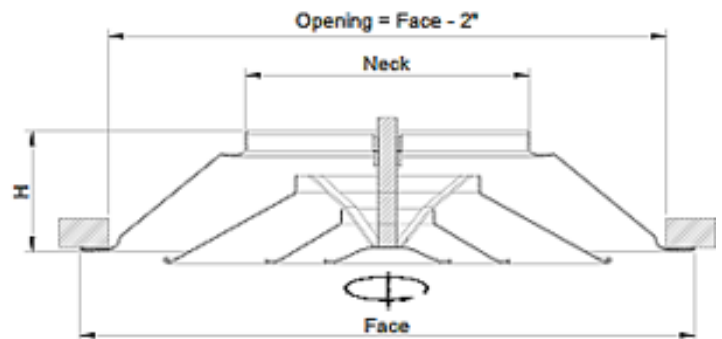
BFD



Types: We have 2 types of RCD.

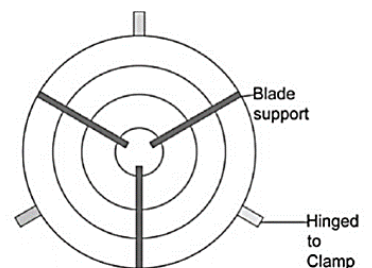
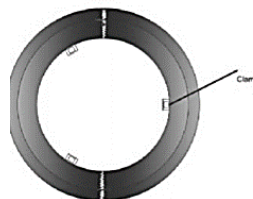
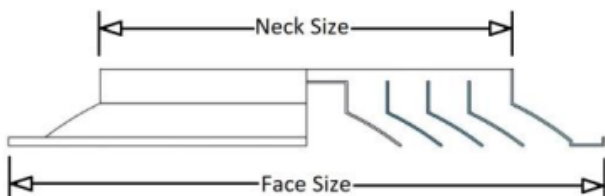
- **RCD with Adjustable Core.**

The Concentric rings are connected on a central screw. Adjust the inner ring using a screwdriver to Control the air flow.



- **RCD with fixed Core.**

Each ring is independently fixed to a shared blade support, restricting their movement.



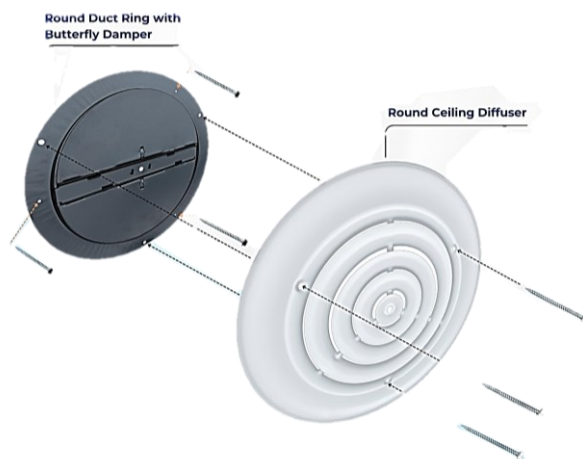
Performance

NECK SIZE (inches)	NECK AREA (Sq. Ft.)	NECK VELOCITY (FPM)	300	400	500	600	700	800	900	1000	1100	1200	1400	1600
		Velocity Pressure	0.006	0.010	0.016	0.023	0.031	0.040	0.051	0.063	0.075	0.090	0.122	0.160
6 Dia.	0.196	Air Flow. CFM	60	80	100	120	135	155	175	195	215	235	275	315
		Total Pressure												
		H	0.024	0.041	0.065	0.093	0.126	0.162	0.207	0.255	0.304	0.365	0.494	0.648
		V	0.035	0.058	0.093	0.133	0.180	0.232	0.296	0.365	0.435	0.522	0.708	0.928
		Radius of Diff., Feet	2-3	2-4	3-5	3-5	4-6	4-6	5-7	5-7	5-7	6-8	6-8	7-9
		NC	<20	<20	<20	<20	<20	23	26	30	33	35	40	44
8 Dia.	0.349	Air Flow. CFM	105	140	175	210	245	280	315	350	385	420	490	560
		Total Pressure												
		H	0.017	0.029	0.046	0.066	0.088	0.114	0.145	0.180	0.214	0.257	0.348	0.456
		V	0.026	0.044	0.070	0.101	0.136	0.176	0.224	0.277	0.330	0.396	0.537	0.704
		Radius of Diff., Feet	2-4	3-5	4-6	4-7	5-7	6-8	6-8	7-9	7-10	7-10	8-11	9-12
		NC	<20	<20	<20	21	26	31	34	37	40	44	48	>50
10 Dia.	0.545	Air Flow. CFM	165	220	275	325	380	435	490	545	600	655	765	870
		Total Pressure												
		H	0.017	0.028	0.044	0.063	0.085	0.110	0.140	0.173	0.206	0.248	0.335	0.44
		V	0.025	0.041	0.066	0.095	0.128	0.165	0.210	0.260	0.309	0.371	0.503	0.659
		Radius of Diff., Feet	3-6	5-7	5-8	6-9	7-9	8-10	8-11	9-12	9-12	10-13	11-14	12-15
		NC	<20	<20	<20	<20	21	26	30	33	36	39	44	48

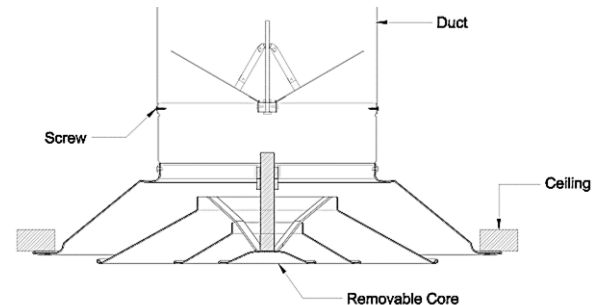
Installation

Round diffusers have some key differences from square ones. They can't be installed with ceiling tiles using lay-in methods; they need a plaster frame. They also don't require a plenum box and usually connect directly to flexible ducts.

- ♦ Fixing directly to duct by screws.
- ♦ By Help of T-Ring (By Other).
- ♦ 3-Points Fixing
(By Screwing to the Face or by Mounting Bracket).



- ♦ 3-Points Fixing by Screwing to the Face



- ♦ Fixing directly to duct by screws.

T-RING (DROP-IN MOUNTING FOR PLASTERED CEILINGS)



- ♦ By Help of T-Ring.

➤ You Can Scan the QR Codes For more Clarification.



Jet Diffusers

Jet diffusers are designed to provide high-velocity air delivery for large spaces and spot cooling. Adjustable throw angle for precise control. They are ideal for applications where long throw is required as in Air Ports or Mosques.

Types:

We have 3 Types of Jet

- **Round Jet:** High velocity air, long distances. Adjustable throw angle. Perfect for spot cooling, large spaces.
- **Nozzle Jet (Eye Ball):** Precise air direction control. Ball-shaped core rotates for targeted airflow. Powerful, versatile.
- **Drum Jet:** Massive air volume, long distances. Adjustable drum and guide for direction and dispersion. Ideal for large spaces such as warehouses, sport halls or factories.

RJD



NJD
(EB)



DJD

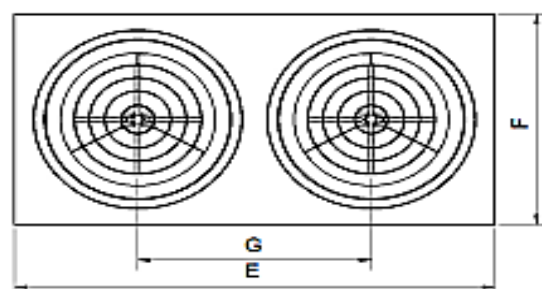


Components:

- **Frame:** The frame is made of aluminum. The face and neck sizes are fixed for each diffuser size.
- **RJD Core:** The core is a series of concentric cylindrical rings that help to accelerate the air stream. The rings can be rotated to direct the airflow.
- **Eye Ball (EB):** Formed Aluminum sheet of fixed sizes can't be modified.
- **Drum:** Holds group-operated vertical Blades in order to enable deflection in horizontal manner.

Options:

- **Multi-element panel for RJD:** Jet diffusers can be ordered as a group of diffusers mounted on a single panel. The customer specifies the number of diffusers and the diffuser size, and the manufacturer calculates the panel dimensions.



Performance:

RJD



SIZE	Neck Velocity (fpm)		300		400		500		600		700		800		1000	
	Velocity Pressure (in. w.g.)		0.006		0.010		0.016		0.022		0.031		0.040		0.062	
	Deflection		0	22.5	0	22.5	0	22.5	0	22.5	0	22.5	0	22.5	0	22.5
8"	Neck Area sq. ft. 0.349	CFM	110		140		180		210		250		280		350	
		SP (in. w.g.)	-	-	0.010	0.020	0.020	0.030	0.024	0.040	0.040	0.060	0.055	0.100	0.090	0.120
		NC	-	-	-	18	-	18	15	18	20	21	22	26	25	32
		Throw (ft.)	14	12	18	17	20	18	22	20	24	21	26	23	28	25
10"	Neck Area sq. ft. 0.545	CFM	170		220		280		330		390		440		550	
		SP (in. w.g.)	-	-	-	0.020	0.010	0.030	0.018	0.040	0.030	0.050	0.040	0.062	0.054	0.091
		NC	-	-	-	-	-	15	-	20	20	22	15	16	24	18
		Throw (ft.)	21	18	24	20	26	21	28	24	30	27	32	28	34	31
12"	Neck Area sq. ft. 0.785	CFM	240		320		400		480		550		630		790	
		SP (in. w.g.)	-	-	-	0.010	-	0.025	0.035	0.040	0.040	0.050	0.050	0.100	0.050	0.090
		NC	-	-	-	-	-	-	-	-	-	20	18	22	22	32
		Throw (ft.)	25	21	27	23	28	26	31	28	34	31	35	32	41	38



Note: If the de flection angle not mentioned consider it at 0°.

NJD (EB)



Nominal Size (mm)	Effective Area (m2)	Air Volume (CFM)	Pressure Drop (Pa)	Noise dB(A)	Throw (m)
150	0.005	59	18.5	26	10.8
		74	22.8	30	13.5
		95	56	34	17.4
		118	82	39	22.2
		148	116	43	27.3
200	0.009	95	10	30	12.7
		118	21	33	16
		148	54	38	20
		189	82	41	25.7
		236	116	45	32.2
250	0.0145	148	11	29	12.9
		189	22	34	16.9
		236	55	39	25.2
		296	81	42	31.5
		372	116	46	37.5

DJD

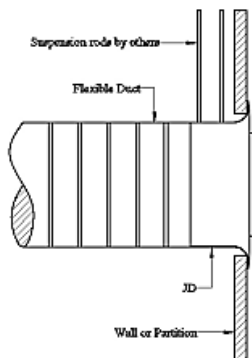



Model	CFM	Air Volume							
		296	372	414	473	532	591	739	946
DJD A	Throw (ft)	21-44	26-55	29-62	34-71	38-80	43-89	53-110	67-123
	Ps	25.1	39.1	49.5	64.3	81.3	100	156	256
	dBA	32	34	36.5	40	42	44	47.5	51
DJD B	Throw (ft)	-	25-52	28-57	31-66	32-75	39-83	49-107	64-120
	Ps	-	30.8	45.7	51	59	76	142	180
	dBA	-	33.5	35.4	38	40	42	46.5	50
DJD C	Throw (ft)	-	-	-	21-45	24-51	27-56	34-70	43-90
	Ps	-	-	-	20	25	31.8	70	108
	dBA	-	-	-	34	35	36	40.5	45
DJD D	Throw (ft)	-	-	-	-	19-40	21-45	27-56	34-73
	Ps	-	-	-	-	<10	<20	30	61
	dBA	-	-	-	-	26	30.3	37	41
DJD E	Throw (ft)	-	-	-	-	-	-	-	28-59
	Ps	-	-	-	-	-	-	-	26
	dBA	-	-	-	-	-	-	-	37.5

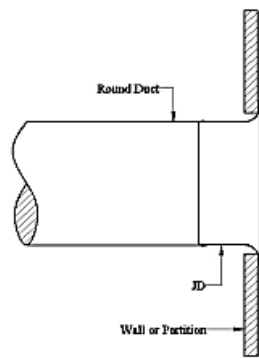
Installation:

RJD Installation

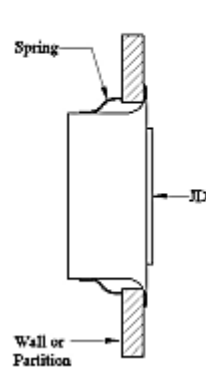
RJD With Flexible duct connection (Option F)



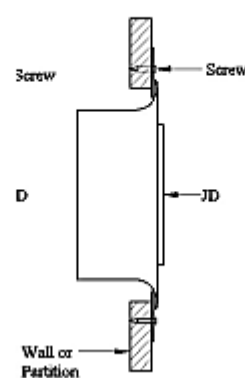
RJD With Round duct connection (Option R)



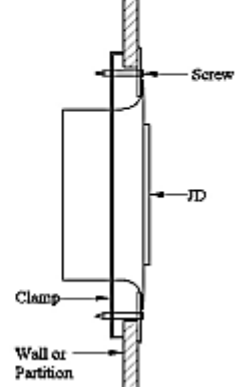
RJD Fixed by Spring (Option Sp)



RJD Multi elements Fixed by Screw (Option S)



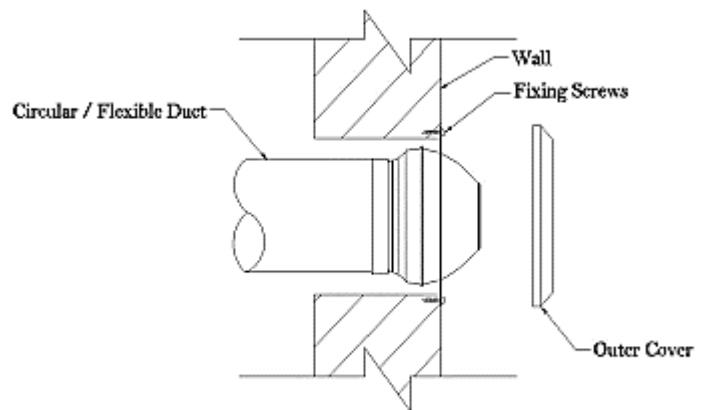
RJD Multi elements Fixed by Clamp (Option C)



Duct Connection: By Screws or
Suspension Rod (By Other)

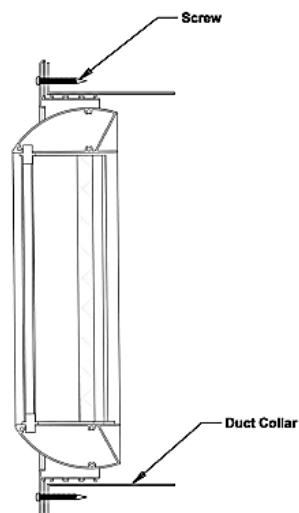
Wall Partition: By Spring Clip or Fixing Screws or Clamp & Screw

NJD (EB) Installation

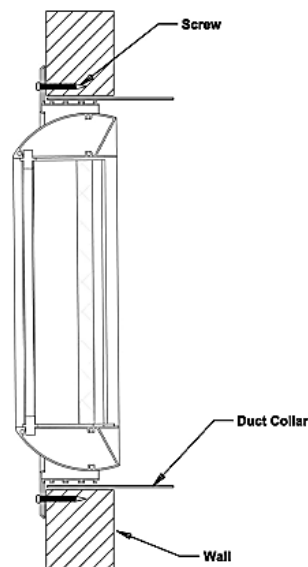


DJD Installation

Option A: By Screw
and collar (Duct)



Option B: By Screw
(Wall mounted)



Disc valve:

Disc valves are typically used for exhaust applications in bathrooms and kitchens. They do not have a core or obstruction damper, and they have a simple, circular design.



Components:

- **Material:** High quality steel and polyester powder coated.
- **Frame:** The frame is similar to the frame of a jet diffuser.
- **Core:** The core is a disc that is shaped like a shallow dish. The dish is concave facing the air flow, which helps to maintain airflow uniformity. The disc is attached to the frame with a screw, and the height of the disc can be adjusted by adjusting the screw from the face to control the airflow rate.

Options:

- **Three sizes are available: 4, 6, and 8 inches. Each size has performance tables for each position of the disc.**

Performance:

- First You have to Select the Correct table according to the required size, then you can determine its performance as clarified.

Adjustments by rotating the Screw →

Size 6" ↓

Setting	Air Flow (CFM)	30	35	40	50	60	70	80	90	100	120	140	160	180
-1/4"	SP	0.3	0.5	0.6	1.4									
	NC	<20	21	24	32									
-1/8"	SP	-	0.2	0.3	0.4	0.6	0.8	1.1						
	NC	-	<20	<20	21	25	31	34						
0	SP	-	-	-	0.2	0.3	0.4	0.6	0.8	1				
	NC	-	-	-	<20	20	23	26	34	40				
1/4"	SP	-	-	-	-	-	-	0.2	0.3	0.4	0.6	0.8	1.1	
	NC	-	-	-	-	-	-	<20	<20	22	31	35	42	
1/2"	SP	-	-	-	-	-	-	-	0.2	0.3	0.4	0.6	0.8	1
	NC	-	-	-	-	-	-	-	<20	<20	24	28	35	45

Installation:

Note: Standard setting consideration it at 0°.

- You can Scan the QR Code for more clarification

